1 Basics of Signals and Systems

Objective

After completing this chapter, the user will be able to:

1. Understand the basics of signal and system
2. Understand Laplace transform and its properties
3. Understand Fourier transform
4. Know the relationship between Laplace transform, and Fourier transform
5. Know how to find the transfer function and impulse response

Introduction

1. Signal is a physical quantity that changes with time, space, and any other variable which is independent in nature.
2. Any fundamental quantity which represents some information is called a signal. Information can be in any form: Analog or Digital.

   E.g.: A speech signal
3. Signals may be proceeded by systems, which may modify the signals.
4. A system is nothing but an entity that proceeds with some set of input and produces some set of outputs.
5. System can be in various forms: Electrical, Mechanical, Physical, etc.

   E.g.: Computer system

Basics of signal:

Unit impulse signal

$x(t) = \delta(t)$

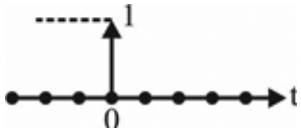


Fig. 1.1

$\delta(t) = 0 \qquad t \neq 0$

$\delta(t) \neq 0 \qquad t = 0$

$t = 0 \rightarrow \delta(t) \rightarrow \infty$

$\int_{-\infty}^{\infty} \delta(t)\,dt = 1$

Impulse signal

$x(t) = A\delta(t)$

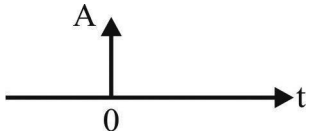


Fig. 1.2

$$\begin{cases} A\delta(t) = 0 & t \neq 0 \\ A\delta(t) \neq 0 & t = 0 \end{cases}$$

$t = 0 \Rightarrow A\delta(t) \rightarrow \infty$

$\int_{-\infty}^{\infty} A\delta(t)\,dt = A\int_{-\infty}^{\infty} \delta(t)\,dt = A$

Note: Impulse signals have a constant area

Unit step signal:

$x(t) = u(t)$

$$u(t) = \begin{cases} 1 & t \geq 0 \\ 0 & t < 0 \end{cases}$$

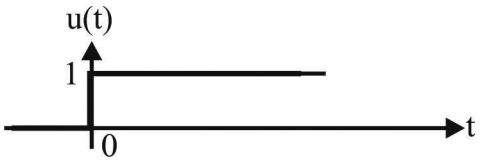


Fig. 1.3

Step signal:

$x(t) = Au(t)$

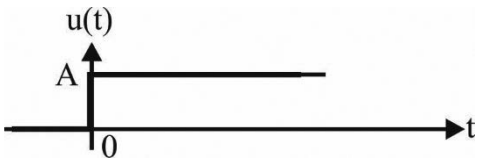


Fig. 1.4

Notwe: Step signals have constant amplitude.

Unit ramp signal

$$x(t) = tu(t)$$

$$x(t) = \begin{cases} t & \geq 0 \\ 0 & t < 0 \end{cases}$$

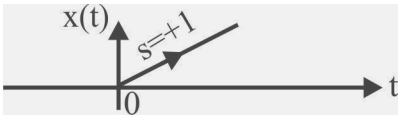


Fig. 1.5

Ramp signal

$$x(t) = Atu(t) = \begin{cases} At & t \geq 0 \\ 0 & t < 0 \end{cases}$$

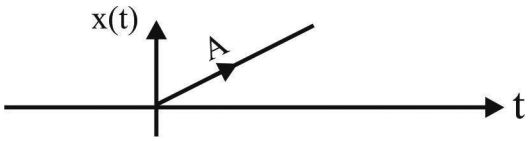


Fig. 1.6

Note: Ramp signals have a constant slope

Parabolic signal

$$p(t) = \begin{cases} \dfrac{At^2u(t)}{2} & t \leq 0 \\ 0 & t > 0 \end{cases}$$

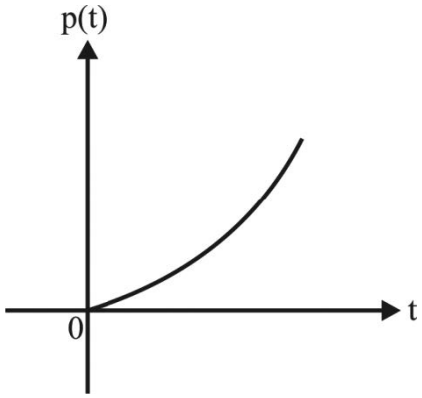


Fig. 1.7

Relation between different signals:

$[\delta(t), u(t) \text{ and } tu(t)]$

Area (vs) Running Integration

$$\int_{-\infty}^{\infty} x(t)dt \rightarrow \text{Area of } x(t)$$

$$\int_{-\infty}^{t} x(t)dt \rightarrow \text{Running Integration of } x(t)$$

$$\frac{d}{dt}(tu(t)) = u(t)$$

$$\frac{du(t)}{dt} = \delta(t)$$

$$\int_{-\infty}^{t} \delta(t)dt = u(t)$$

$$\int_{-\infty}^{t} u(t)dt = tu(t)$$

One sided exponential

$$x(t) = e^{-\alpha t}u(t) \qquad \alpha > 0$$

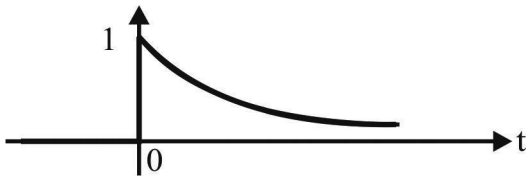


Fig. 1.8

$$x(t) = (e^{\alpha t})u(t) \qquad \alpha > 0$$



Fig. 1.9

$$x(t) = (1 - e^{-\alpha t})u(t) \quad \alpha > 0$$

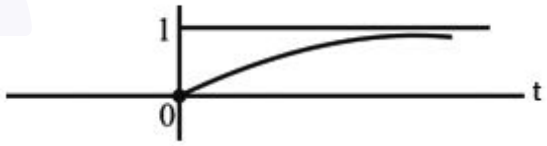


Fig. 1.10

$$A\sin\omega_0 tu(t)$$

Amplitude = A constant

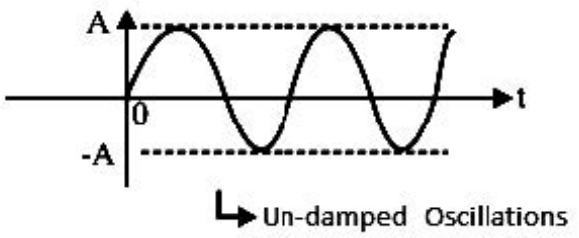


Fig. 1.11

1.2.12

$$A\cos\omega_0 tu(t)$$

Amplitude = A constant

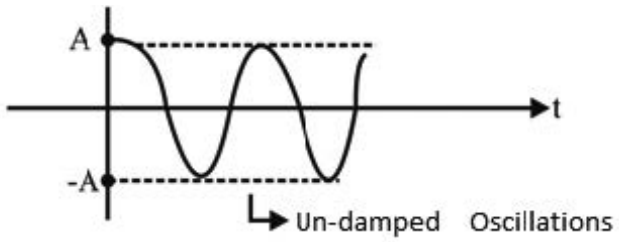


Fig. 1.12

Causal signal

If x(t) = 0 For t < 0

x(t) = u(t) Causal

x(t) = u (t+1) Non-causal

$\delta(t) \rightarrow causal$

$u(t-1) \rightarrow causal$

Procedure to plot

$x(t) = A(t)\cos\omega_0 t \ or \ p(t)\sin\omega_0 t$

Step 1:Plot A(t) or p(t) in Ref form: (dotted)

Step 2:Take a mirror image of A(t) w.r.t Horizontal axis

Step 3:Calculate x(t) at = 0

Step 4:Draw sinusoidal shape in between

Examples

1. $x(t) = t\sin tu(t) = (tu(t))\sin t$

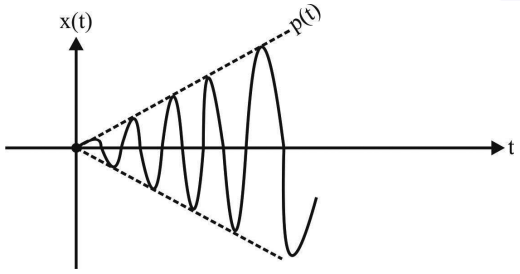


Fig. 1.13

2. $x(t) = e^{-t}\sin t \ u(t) = (e^{-t}u(t))\sin t$

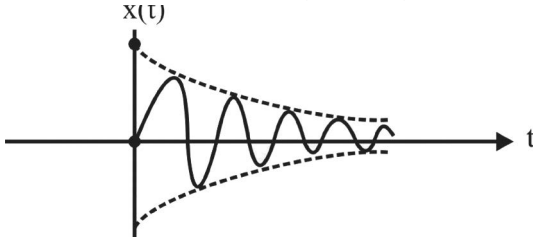


Fig. 1.14

Damped Oscillations

3. $x(t) = e^{2t}\cos t\, u(t)$

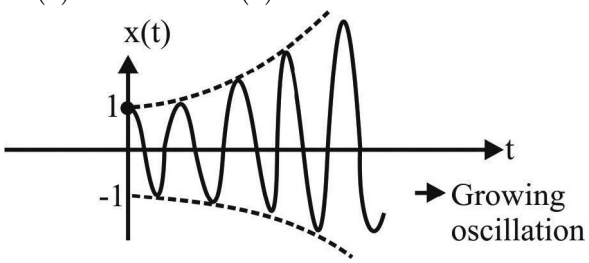


Fig. 1.15

Basic operations on signals

1. Given: x(t) vs t
   Plot: Ax(t) vs t
   Procedure: Multiply vertical axis by A:
2. Given x(t) vs t
   $Plot: x(t-t_0) vs \ t \rightarrow shift \ right \ by \ t_0$
3. Given x(t) vs t
   Plot x(at) vs t
   Divide the horizontal axis by a
4. Given x(t) vs t
   Plot x (-t) vs t
   Mirror image w.r.t. vertical axis

Examples

1. x(t) = u(t - T)

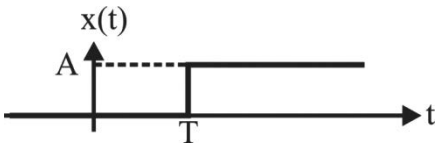


Fig. 1.16

2. x(t) = u(t) –u (t - T)

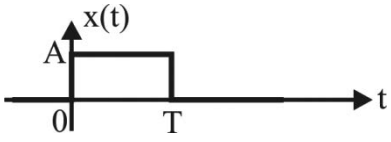


Fig. 1.17

Test yourself:

1. Draw the graph for the following signal.
   $x(t) = (1-e^{-t})u(t)$

Basics of System:

- A system is a mathematical model of a physical process that relates the input (or excitation) signal to the output (or response) signal.

- Let x and y be the input and output signals of a system, respectively.
- Then the system is viewed as a transformation (or mapping) of x into y.
- This transformation is represented by the mathematical notation
  y = Tx.

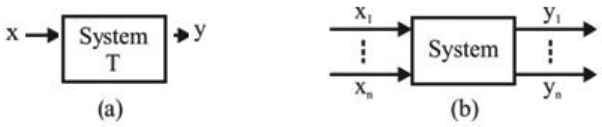


Fig. 1.18

Causal and Non-causal System

- A system is called causal if its output y(t) at an arbitrary time $t = i_0$ depends on only the input, $x(t)$ *for* $t \le t_0$
- That is, the output of the causal system at the present time depends on only the present and/or past values of the input but not on its future values.
- A system is called non-causal if it is not causal. Examples of the non-causal system are

$$y(t) = x(t+1)$$

$$y[n] = x[-n]$$

Point To Remember

All memory-less systems are causal but not vice versa.

Linear Systems and Nonlinear Systems

- A system which satisfies the principle of superposition and homogeneity is called a linear system.
- If the operator T satisfies the following two conditions, then T is called a linear operator and the system represented by a linear operator T is called a linear system:

1. Additivity:

Given that $Tx_1 = y_1$ *and* $Tx_2 = y_2$, *then*

$$T(x_1 + x_2) = y_1 + y_2 \quad (1)$$

for any signals x_1 *and* x_2.

2. Homogeneity (or scaling):

a) $T(\alpha x) = \alpha y$ (2)

for any signals, x and any scalar α.

b) Any system that does not satisfy Eq. (1) and (2) is classified as anonlinear system.

c) Equations (1) and (2)can be combined into a single condition as

$$T(\alpha_1 x_1 + \alpha_2 x_2) = \alpha_1 y_1 + \alpha_2 y_2 \quad (3)$$

where α_1 *and* α_2 are arbitrary scalars.

d) Equation (3) is known as the superposition property.

e) Examples of linear systems are the resistor and the capacitor.

Examples of nonlinear systems are:

$$y = x^2$$

$$y = \cos x$$

Time-invariant and time-varying systems:

1. A system is called time-invariant if a time shift (delay or advance) in the input signal causes the same time shift in the output signal. Thus, for a continuous-time system, the system is time-invariant if

$$T\{x(t-\tau)\} = y(t-\tau) \quad (4)$$

for any real value of τ

2. For a discrete-time system, the system is time-invariant (or shift-invariant) if

$$T\{x[n-k]\} = y[n-k] \quad (5)$$

for any integer k.

3. A system which does not satisfy Eq. 4(continuous-time system) or Eq. 5 (discrete-time system) is called a time-varying system.

Linear time-invariant systems:

A system which is both linear and time-invariant is called as Linear Time-Invariant (LTI) system.

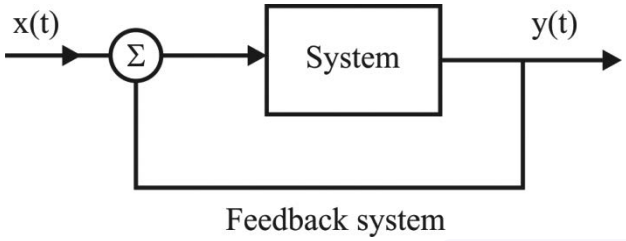
Stable systems

A system is said to be Bounded Input Bounded Output (BIBO) stable if for any bounded input

x defined by $|x| \le k_1$ the corresponding output y defined by $|y| \le k_2$ is also bounded, where k_1 *and* k_2 are finite real constants.

Feedback Systems

1. A special class of systems of great importance consists of systems with feedback.
2. In a feedback system, the output signal is feedback and added to the input of the system as shown in the figure.



Feedback system

Fig. 1.19

Examples

1. Check the linearity of the following system.

   $y = T(x) = x^2$

Solution:

$$T(x_1 + x_2) = (x_1 + x_2)^2 = x_1^2 + x_2^2 + 2x_1x_2$$

$$\neq T\{x_1\} + T(x_2) = x_1^2 + x_2^2$$

Thus, the system is nonlinear.

Laplace transform:

Definition of laplace transform:

1. It is a mathematical tool which is used to solve ordinary differential equations.
2. Solving differential equations in time domain, therefore, reduces to solving algebraic equations in the s-domain.
3. Having done the latter (last) for the desired unknowns, their values as functions of time may be found by taking inverse transforms.
4. Another advantage of using the Laplace transform for solving differential equations is that initial conditions play an important role in the transformation process, so they are automatically incorporated into the solution.
5. The mathematical expression for Laplace transform is,

   $L[f(t)] = F(s), t \ge 0$

   where, $F(s) = \int_0^\infty f(t).e^{-st}dt$
6. The original time function f(t) is obtained back from the Laplace transform by a process called inverse Laplace transformation and denoted as L^{-1}

   Thus, $L^{-1}[L(f(t))] = L^{-1}[F(s)] = f(t)$

Do you know?

Laplace transform has two types: Unilateral (one sided) and Bilateral (two sided)

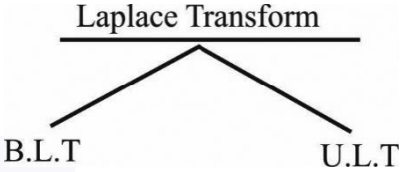


Fig. 1.20

$$x(s) = \int_{-\infty}^{\infty} x(t)e^{-st}dt \qquad x(s) = \int_{0^-}^{\infty} x(t)e^{-st}dt$$

Note: Lower limit is 0^- to include impulses

Properties of uni lateral transform (ULT):

1. Shifting in the time domain

   $x(t) \longleftrightarrow x(s)$

   $x(t - t_0) \longleftrightarrow e^{-st_0}x(s)$

   $x(t + t_0) \longleftrightarrow e^{st_0}x(s)$
2. Shifting in the S domain

   $x(t) \longleftrightarrow x(s)$

   $e^{s_0 t}x(t) \longleftrightarrow x(s - s_0)$

   $e^{-s_0 t}x(t) \longleftrightarrow x(s + s_0)$
3. Multiplication Property

   $x(t) \longleftrightarrow x(s)$

   $tx(t) \longleftrightarrow \frac{-dx(s)}{ds}$
4. Time differentiation property

   $x(t) \longleftrightarrow x(s)$

$$\frac{dx(t)}{dt} \longleftrightarrow sx(s) - x(0^-)$$

$$\frac{d^2x(t)}{dt^2} \longleftrightarrow s\{sx(s) - x(0^-)\} - \left\{\frac{dx(t)}{dt}\right\}_{t=0^-}$$

$$= s^2x(s) - sx(0^-) - (dx(t)/dt)_{t=0^-}$$

$$= \frac{d^3x(t)}{dt^3} \longleftrightarrow s\left\{\begin{matrix} s^2x(s) - sx(0^-) \\ -\left(\frac{dx(t)}{dt}\right)_{t=0^-} \end{matrix}\right\} - \left(\frac{d^2x(t)}{dt^2}\right)_{t=0^-}$$

Initial condition (vs)initial value:

$x(0^-)$ $\qquad$ $x(0^+)$

$\left(\frac{dx(t)}{dt}\right)_{t=0^-}$ $\qquad$ $\left(\frac{dx(t)}{dt}\right)_{t=0^+}$

$$x(t) \longleftrightarrow x(s)$$

$$\int_{-\infty}^{t} x(t)dt \longleftrightarrow \frac{x(s)}{s}$$

Initial value theorem (IVT):

$$\underset{t\to 0}{Lim}\ x(t) = \underset{s\to\infty}{Lim}\ sX(s)$$

Limitations of I.V.T:

1. IVT is valid only when x(t) is causal
2. IVT is valid for strictly proper functions (Order of denominator polynomial is greater than numerator polynomial)

$$x(s) = \frac{N(s)}{D(s)} \qquad Dr > Nr$$

Final value theorem (FVT):

$\underset{t\to\infty}{Lim}\ x(t)$ = Final value or steady-state value of x(t)

$$\lim_{t\to\infty} x(t) = \lim_{s\to 0} s \times (s)$$

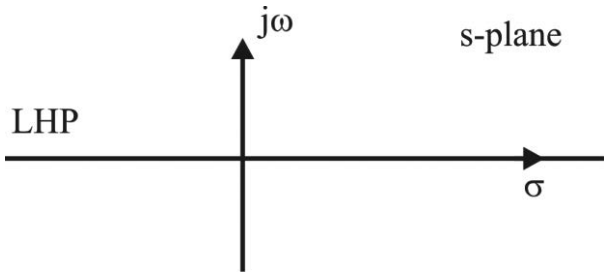


Fig. 1.21

Limitations of FVT:

1. FVT is valid only when x(t) is causal
2. All the poles of sX(s) should be STRICTLY in theleft half of the s-plane (LHP)

Case 1: If all the poles of sX (s) lie in LHP, then

1. FVT is applicable
2. FV is finite

Case 2: If pole locations of sX(s) are as shown in the figures, then

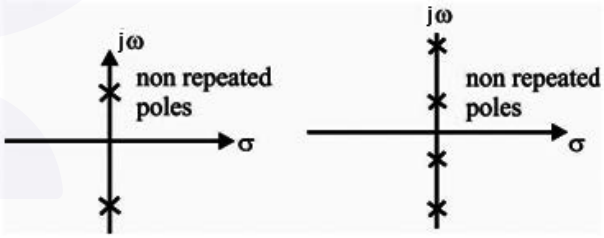


Fig. 1.22

1. FVT is not applicable
2. FVT is indeterminate value

Case 3: Remaining cases as shown in below figures:

1. FVT is not applicable
2. $FV \to \infty$

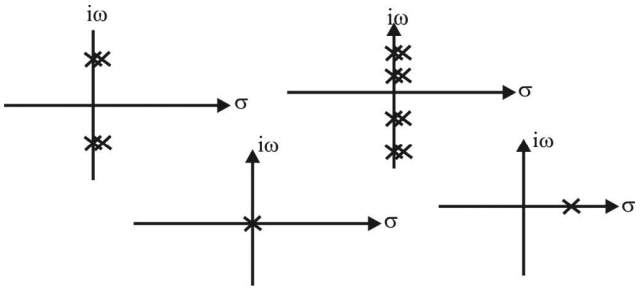


Fig. 1.23

Indeterminate (vs) undefined

$Lim \sin t \Rightarrow -1 \le x(t) \le 1$
$t \to \infty$

$\left(\delta(t)\right)_{t=0} \to +\infty$

Finite but uncertain undefined

$x(t) \xleftrightarrow{L.T.} x(s)$

s : *complex* variable

$x(s)$: *s domain signal*

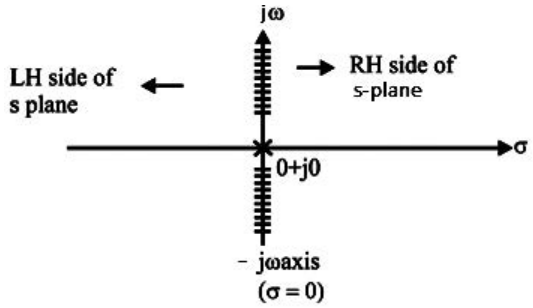


Fig. 1.24

Case 1: $x(s) = \dfrac{(s+1)}{(s-2)(s+3)} = \dfrac{N(s)}{D(s)}$

Zeros: N(s) = 0 $\to s = -1$

$poles : D(s) = 0 \to s = 2, s = -3$

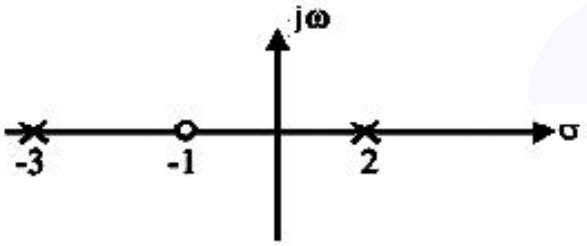


Fig. 1.25

Case 2

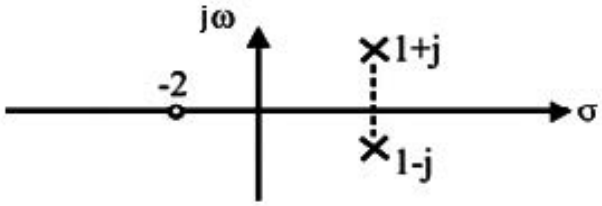


Fig. 1.26

$$x(s) = \frac{k\left(s-(-2)\right)}{\left(s-(1+i)\right)\left(s-(1-(i)\right)}$$

Laplace transform of basic elementary functions:

S.No.	f(t)	F(s) = L[f(t)
1.	$\delta(t)$ unit impulse at t = 0	1
2.	u(t) unit step at t = 0	$\frac{1}{s}$
3.	u(t –T) unit step at t = T	$\frac{1}{s}e^{-sT}$
4.	t	$\frac{1}{s^2}$
5.	$\frac{t^2}{2}$	$\frac{1}{s^3}$
6.	t^n	$\frac{n!}{s^{n+1}}$
7.	e^{at}	$\frac{1}{s-a}$
8.	e^{-at}	$\frac{1}{s+a}$
9.	te^{at}	$\frac{1}{(s-a)^2}$
10.	te^{-at}	$\frac{1}{(s+a)^2}$
11.	$t^n e^{-at}$	$\frac{n!}{(s+a)^{n+1}}$
12.	$\sin \omega t$	$\frac{\omega}{s^2+\omega^2}$
13.	$\cos \omega t$	$\frac{s}{s^2+\omega^2}$

Table 1.1 Table of Laplace Transform Pairs

1. Given, $L\left[f(t)\right] = F(s) = \int_0^\infty f(t)e^{-st}dt$, which of the following expressions are correct?

I. $L\left\{f(t-a)u(t-a)\right\} = F(s)e^{-sa}$

II. $L\left\{tf(t)\right\} = \frac{-dF(s)}{ds}$

III. $L\left\{(t-a)f(t)\right\} = as\ F(s)$

IV. $L\left\{\frac{df(t)}{dt}\right\} = sF(s) - f(0^+)$

Select the correct answer using the codes given below.

(A) I, II and III (B) I, II and IV

(C) II, II and IV (D) I, III and IV

Solution: (B)

These are the properties of the Laplace transform.

Test yourself

1. If L[f(t)] = F(s), then L[f(t – T)] is equal to

(A) $e^{sT}F(s)$ (B) $e^{-sT}F(s)$

(C) $\frac{F(s)}{1+e^{sT}}$ (D) $\frac{F(s)}{1-e^{-sT}}$

Solution: (B)

2. $F(s) = \frac{\left(1-e^{-sT}\right)}{s}$ is the Laplace transform of

(A) a pulse of width T

(B) a square wave of period T

(C) a unit step delayed by T

(D) a ramp delayed by T

Solution: (A)

Relationship between Laplace and Fourier transform:

1. Fourier transform is a special case of Laplace transform.
2. Consider σ asthe real axis and jω as the imaginary axis indicator.
3. In the complex domain, if you evaluate Laplace transform only on the imaginary axis, you will get Fourier transform.

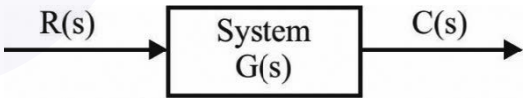
The term 's' in Laplace domain is s = σ +jω, if σ=0, Laplace transform is a Fourier transform.

4. The Laplace transform evaluated at s=jω is equal to the Fourier transform if its region of convergence (ROC) contains the imaginary axis.

Transfer function:

Definition of transfer function:

1. For a control system, there exists an input which is termed as excitation and produces an output called as response of the system.
2. This cause and effect relationship is best illustrated by means of a transfer function.
3. Transfer function is the ratio of Laplace transform of response to Laplace transform of excitation, assuming all initial conditions are zero.
4. If we express in terms of Laplace transform



G(s) = Transfer Function = $\frac{C(s)}{R(s)}$

The arrow indicates the direction of control action; that is, by the application of excitation (R(s)), the response (C(s)) of the system is controlled.

Mathematical representation of LTI systems:

1. Unit Impulse Response of LTI S/S h(t)
2. $h(t) \xleftrightarrow{L.T.} H(s)$
3. $h(t) \xleftrightarrow{L.T.} H(\omega)$

Consider RC Low pass Network:

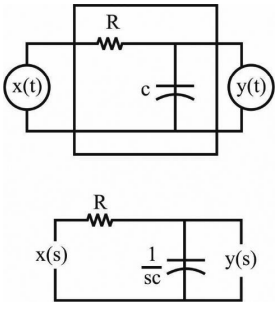


Fig. 1.27

$$\frac{Y(s)}{X(s)} = \frac{1/SC}{R + \frac{1}{SC}} = \frac{1}{1 + SRC}$$

$$\frac{Y(s)}{X(s)} \rightarrow \frac{L.T.\left(y(t)\right)}{L.T.\left(x(t)\right)} = H(s) = \frac{1}{1 + SRC}$$

H(s)is Transfer function of LTI system

$$H(s) = \frac{1/RC}{S + \frac{1}{RC}}$$

$$h(t) = \frac{1}{RC} e^{-t/RC}$$

Case 1: Transfer function to differential equation

$$H(s) = \frac{Y(s)}{X(s)} = \frac{1}{1 + SCR}$$

$$Y(s) + SRC\ Y(s) = X(s)$$

$$y(t) + RC\frac{dy(t)}{dt} = x(t)$$

$$\frac{dy(t)}{dt} + \frac{1}{RC} y(t) = \frac{1}{RC} x(t)$$

Case 2: Differential equation to Transfer function:

$$\frac{dy(t)}{dt} + \frac{1}{RC} y(t) = \frac{x(t)}{RC}$$

L.T. Both sides

$$sy(s) - y(0^-) + \frac{1}{RC} y(s) = \frac{1}{RC} x(s)$$

$$y(s)\left(s + \frac{1}{RC}\right) = y(0^-) + \frac{x(s)}{RC}$$

$$y(s) = \frac{y(0^-)}{\left(s + \frac{1}{RC}\right)} + \frac{x(s)}{RC\left(s + \frac{1}{RC}\right)}$$

Case A: $y(0^-) = 0 \rightarrow$ Initial conduction of the system.

$$y(s) = \frac{x(s)}{RC\left(s + \frac{1}{RC}\right)}$$

1. $H(s) = \frac{y(s)}{x(s)} = \frac{1/RC}{\left(s + \frac{1}{RC}\right)}$
2. $y(s) = x(s)H(s)$

Case B: $y(0^-) \neq 0$

1. $y(0^-) \neq 0 \qquad H(s) = \frac{y(s)}{x(s)}$
2. $y(s) \neq x(s)H(s)$

Poles and zeros:

1. The transfer function of a linear control system can be expressed in rational form as:

$$G(s) = \frac{k(s - s_1)(s - s_2).....(s - s_n)}{(s - s_a)(s - s_b).....(s - s_m)}$$

where k is gain of the Transfer function

2. If in the transfer function we put $s = s_a, s_b.....,s_m$ the value of the transfer function becomes infinity, and thus these are called as poles of transfer function.
3. If in transfer function we put $s = s_1, s_2.....s_n$, the value of transfer function becomes zero, and these are called as zeros of transfer function.
4. If poles or zeros are complex, then they always appear in complex conjugate pairs.
5. If 2 poles or 2 zeroes coincide, that pole or zero is called as multiple pole or multiple zero.
6. If a pole or zero is single at any location, then it is called as simple pole or simple zero.
7. The number of times a pole or zero is repeated is called as its multiplicity or order.
8. If n > m, then the transfer function goes to infinity as $s \rightarrow \infty$, so we conclude that a pole of the transfer function exists at $s \rightarrow \infty$ of order (n-m).
9. In n < m, then transfer function goes to zero as $s \rightarrow \infty$, so we conclude that

a zero of the transfer function exists at $s \to \infty$ of order (m-n)

10. Therefore, for a rational transfer function, the total number of poles is equal to the total number of zeroes

$$G(s) = \frac{(s+2)}{(s+1)(s^2+9)}$$

The transfer function has one zero at s = -2 and 3 poles at s = -1, s = ±3j. But since the number of poles must be equal to the number of zeroes, so there are 2 zeroes at infinity.

Note: Types of transfer functions

1. The transfer functions are said to be strictly proper if the order of the denominator polynomial is greater than that of the numerator polynomial (i.e. m > n). If m = n, the transfer function is called proper. The transfer function is improper if n > m.
2. In the transfer function expression of a control system, the highest power of s in the numerator is generally either equal to or less than that of the denominator.

Unit impulse response:

$$r(t) = \begin{cases} \infty, & t = 0 \\ 0, & t \neq 0 \end{cases} \quad : \quad \int_{-\infty}^{+\infty} \delta(t)\,dt = 1$$

Laplace transform of R(s) = 1

Transfer function, $G(s) = \frac{C(s)}{R(s)}$

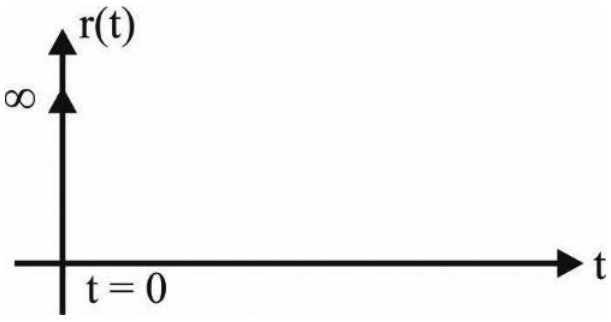


Fig. 1.28

Let,

C(s) = G(s) R (s)

R(s) = Impulse signal = 1

C(s) = Impulse response = G(s) × 1 = T.F.

L[Unit Impulse Response] = Transfer function

$$= \left[\frac{C(s)}{R(s)}\right]$$

Response of control system:

$$Transfer\ function, G(s) = \frac{C(s)}{R(s)}$$

C(s) = G(s).R(s)

1. If impulse input is applied, then $r(t) = \delta(t)$
2. Laplace Transform of excitation, R(s) = 1
3. The response of the system to the impulse input is termed as Impulse response C(s) = G(s)

   Which means that the Laplace transform of unit impulse response with initial conditions are assumed to be zero is called Transfer Function.

1. Consider, a linear time-invariant system has the input u(t) and output y(t). The system can be characterized by its impulse response g(t), which is defined as the output when the input is a unit-impulse function $\delta(t)$.
2. Once the impulse response of a linear system is known, the output of the system y(t), with any input u(t), can be found by using the transfer function.
3. Let G(s) denote the transfer function of a system with input u(t). Output y(t), and impulse response g(t). The transfer function G(s) is defined as

$$G(s) = L[g(t)] = \frac{L[y(t)]}{L[u(t)]}_{initial\ conditions \to 0}$$

$$= \frac{Y(s)}{U(s)}$$

Common mistake:

Sometimes students do a common mistake; they first find y(t)/u(t) and then take its Laplace transform to determine the transfer function, which is absolutely wrong. Because,

$$G(s)=\frac{Y(s)}{U(s)}=\frac{L\left[y(t)\right]}{L\left[u(t)\right]}\neq L\left[\frac{y(t)}{u(t)}\right]$$

Points To Remember

1. The relation between the responses of the system to the singularity function is the same as the relation between the singularity function themselves.
2. $\frac{d}{dt}$ (Parabolic Response) = Ramp response
3. $\frac{d}{dt}$ (Ramp Response) = Step response
4. $\frac{d}{dt}$ (Step Response) = Impulse response

1. Weare deriving the transfer function of a linear element with

(A) Both initial conditions and loading are taken into account

(B) Initial conditions are taken into account, but the element is assumed to be not loaded

(C) Initial conditions are assumed to be zero, but loading is taken into account

(D) Initial conditions are assumed to be zero, and the element is assumed to be not loaded.

Solution: (C)

While deriving the transfer function of a linear element, only initial conditions are assumed to be zero, and loading (or input) can't be assumed to be zero.

Properties and limitations of the transfer function (TF)

The properties of the transfer function are summarised as follows:

1. The transfer function is defined only for a linear time-invariant system. It is not defined for non-linear or time-variant systems.
2. The transfer function between an input variable and an output variable of a system is defined as the Laplace transform of the unit impulse response with initial conditions are assumed to be zero. Alternately, the transfer function between a pair of input and output variables is the ratio of the Laplace transform of the output to the Laplace transform of the input with initial conditions are assumed to be zero.
3. All initial conditions of the system are set to zero.
4. Transfer function is independent of the input of the system.
5. The transfer function of a continuous data system is expressed only as a function of complex variables. It is not a function of the real variable, time or any other variable that is used as the independent variable or discrete-data system modelled by difference equations; the transfer function is a function of Z when the Z-transform is used.
6. If the system transfer function has no poles or zeros with positive real parts, the system is a minimum phase system.

   Non-minimum phase functions are the functions which have poles or zeros on right hand side of s-plane.
7. The stability of a time-invariant linear system can be determined from its characteristic equation.

   Characteristic equation: The characteristic equation of an L.T.I. system is defined as the equation obtained by setting the denominator polynomial of the closed-loop transfer function to zero.

How to calculate response of a system:

If Initial conditions are zero:

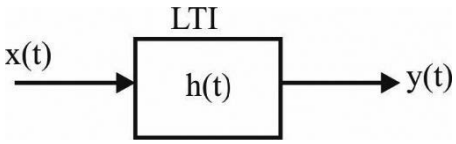


Fig.

1. Transfer function can be calculated:

$$H(s)=\left\{\frac{Y(s)}{X(s)}\right\}_{IC=0}$$

2. Output can be calculated by using TF

   y(s) = x(s) H (s)

   $y(t)=L^{-1}(Y(s))$

3. $y(t)=x(t)*h(t)$

   If Initial conditions are not zero:

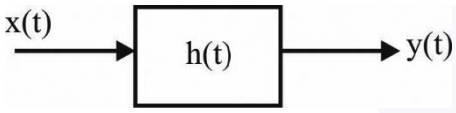


Fig.

1. Transfer function can be calculated.

$$H(s)=\left\{\frac{y(s)}{x(s)}\right\}_{IC=0}$$

2. Output can be calculated by using T.F.

   $y(s)\neq x(s)H(s)$

3. $y(t)\neq x(t)*h(t)$

   $H(s)=\dfrac{y(s)}{x(s)}\rightarrow$ *Form Differential* Equation

   Use Unilateral Laplace transform and incorporate initial conditions

   Rearrange the equation to get Y(s)

   Apply Inverse Laplace transform to get response y(t).

1. The Transfer function of the LTI system is given as $H(s)=\dfrac{s-2}{s+3}$. Calculate y(t), if Initial conditions are $y(0^-)=1$ and x(t) = 1.

Solution:

1. $H(s)\rightarrow Given$

2. $IC\neq 0 \qquad Y(s)\neq X(s)H(s)$

$$H(s)=\frac{(s-2)}{(s+3)}=\frac{Y(s)}{X(s)}$$

$$sY(s)+3Y(s)=sX(s)-2X(s)$$

$$\frac{dy(t)}{dt}+3y(t)=\frac{dx(t)}{dt}-2x(t)$$

$$x(t)=1.u(t)$$

$$x(0^-)=0$$

$$sY(s)-y(0^-)+3Y(s)$$
$$=sX(s)-X(0^-)-2X(s)$$

$$sY(s)-1+3Y(s)=sX(s)-2X(s)$$

$$Y(s)(s+3)=1+X(s)(s-2)$$

$$Y(s)=\frac{1}{(s+3)}+\frac{1(s-2)}{s(s+3)}$$

$$Y(s)=\frac{1}{(s+3)}+\frac{(s-2)}{s(s+3)}$$

$$=\frac{2s-2}{s(s+3)}=\frac{a_0}{s}+\frac{a_1}{(s+3)}$$

$$=\frac{-2}{3s}+\frac{8/3}{(s+3)}$$

$$y(t)=\frac{-2}{3}+\frac{8}{3}e^{-3t} \text{ for } t\geq 0^-$$

Exercise Problems

1. When a human being tries to approach an object, his brain acts as: **[ESE-1999]**

   (A) An error measuring device

   (B) A controller

   (C) An actuator

   (D) An amplifier

Solution: (B)

2. Which one of the following pairs is NOT correctly matched? (Input x(t) and output y(t)) **[ESE-1999]**

(A) Unstable system $\frac{dy(t)}{dt} - 0.1y(t) = x(t)$

(B) Nonlinear system

$$\frac{dy(t)}{dt} + 2t^2 y(t) = x(t)$$

(C) Non causal system y(t) = x(t + 2)

(D) Non dynamic system $y(t) = 3x^2(t)$

Solution: (B)

3. A system is characterised by the input-output relation:

$$y(t) = x(2t) + x(3t)$$

For all t, where y(t) is the output and x(t) is the input, it is **[ESE-2014]**

(A) linear and causal

(B) linear and non-causal

(C) non-linear and causal

(D) non-linear and non-causal

Solution: (B)

4. The ramp function can be obtained from the unit impulse at t = 0 by **[ESE-2013]**

(A) Differentiating unit impulse function once

(B) Differentiating unit impulse function twice

(C) Integrating unit impulse function once

(D) Integrating unit impulse function twice

Solution: (D)

5. If v-i characteristic of a circuit is given by $v(t) = t.i(t) + 2$, the circuit is of which type? **[ESE-2008]**

(A) Linear and time-invariant

(B) Linear and time-variant

(C) Non linear and time-invariant

(D) Non linear and time-variant

Solution: (D)

Exercise Problems

1. (B)

2. (B)

a) For the system to be stable, the roots of the characteristic equation should lie in the LHS of s–plane.

$$\frac{dy(t)}{dt} - 0.1y(t) = x(t)$$

$$\Rightarrow \frac{Y(s)}{X(s)} = \frac{1}{s - 0.1}$$

$\Rightarrow$ $s = 0.1$ which is in RHS of s–plane; hence the system is unstable.

b) The system is linear if the response to $ax_1(t) + bx_2(t)$ is $ay_1(t) + by_2(t)$

$$\frac{dy(t)}{dt} + 2t^2 y(t) = x(t)$$

The condition of linearity is satisfied here; hence the system is linear.

c) A system is causal if the output at any time depends only on values of the input at the present time and in the past.

d) A system is said to be memoryless or static if its output for each value of the independent variable at a given time is dependent only on the input at the same time.

3. (B)

Given $y(t) = x(2t) + x(3t)$

Scaling operation is always a non-causal operation.

Now check for linearity

$$y_1(t) = x_1(2t) + x_2(3t)$$

Also $y_2(t) = x_2(2t) + x_2(3t)$

$$y_3(t) = x_3(2t) + x_3(3t) \quad \text{...(i)}$$

Where

$$x_3(t) = x_1(t) + x_2(t) \quad \text{...(ii)}$$

From (i) and (ii)

$$y_3(t) = x_1(2t) + x_2(2t) + x_1(3t) + x_2(3t)$$

$$y_3(t) = y_1(t) + y_2(t)$$

Thus, the given system is a linear and non-causal system.

4. (D)

$$y = \iint \delta(t)dt$$

$$= t\,u(t) = \text{ramp function}$$

5. (D)

$$v(t) = t\ i(t) + 2$$

$$i_1(t) \rightarrow v_1(t) = t\ i_1(t) + 2$$

$$i_2(t) \rightarrow v_2(t) = t\ i_2(t) + 2$$

Let, $i_3(t) \rightarrow A\ i_1(t) + B\ i_2(t)$

Then,

$$v_3(t) = t\ i_3(t) + 2$$

$$= t\left[A\ i_1(t) + B\ i_2(t)\right] + 2$$

$$= At\, i_1(t) + Bt\ i_2(t) + 2$$

$$\neq Av_1(t) + Bv_2(t)$$

Therefore, the given circuit is non-linear.

Now let $i_2(t) = i_1\left(t - t_o\right)$

So $v_2(t) = t\ i_2(t) + 2$

$$= t\ i_1\left(t - t_0\right) + 2$$

$$\neq \left(t - t_0\right) i_1\left(t - t_0\right) + 2$$

$$\Rightarrow v_2(t) \neq v_1\left(t - t_0\right)$$

Therefore, the given circuit is a time-variant.

2 Block Diagram Representation of a System

Objective

After completing this chapter, the user will be able to understand

- Basics of a block diagram
- Open-loop control system
- Concept of DC gain
- Closed-loop control system and its block diagram
- Difference between open-loop and closed-loop control system
- Block diagram reduction techniques

Introduction

- Any system which controls the value of a physical variable is called a control system. Example: Heating and air conditioning systems
- Control: It means to regulate, direct or command a system such that a desired response is obtained.
- We all have a control system surrounding us.
- Block diagram is helpful in describing a control system.

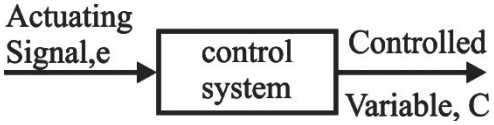


Fig. 2.1

- Input signal is treated as actuating signal, and the output of the control system is known as the controlled variable.
- Control system can be with or without feedback.
- Block diagram is helpful in finding a transfer function of a control system.

Block Diagram

- A control system is composed of many sub-systems, and each sub-system can be represented in terms of its Transfer Function as building blocks of the system.
- These blocks are interconnected in a certain manner to work together and complete the system.
- When we represent the system as an interconnection of these blocks, it results in a block diagram representation of the system.
- Block diagram is a graphical representation of functions performed by individual components and the flow of signals from one component to another component.
- For example, we can consider an automobile which consists of several sub-systems like accelerator, steering wheel, wheels, brakes, etc.
- The transfer functions of the components are usually entered in the corresponding blocks, which are connected by arrows to indicate the direction of the flow of signals.
- Note that the signal can pass only in the direction of the arrows. Thus, a block diagram of a control system explicitly shows a unilateral property.

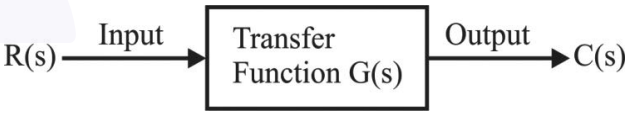


Fig. 2.2

$$C(s) = R(s).G(s)$$

$$G(s) = \frac{C(s)}{R(s)}$$

Note: The dimension of the output signal from the block is the dimension of the input signal multiplied by the dimension of the transfer function in the block.

Open Loop Control System (O.L.C.S.)

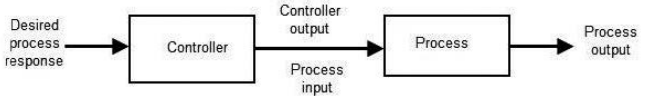


Fig. 2.3

If there is no feedback from the output C(s) to the input R(s), then the system is said to be an Open Loop Control System.

∴ The output of O.L.C.S. is affected not only by input variations but also by a disturbance on it. Open-loop control system doesn't automatically correct the disturbance.

Accuracy of O.L.C.S. depends on the accuracy of input calibration.

Transfer function of O.L.C.S. is $\frac{C(s)}{R(s)} = G(s)$

Some common examples of an open-loop control system are

1. Traffic light controller operated based on a timer
2. Electric washing machine
3. Automatic coffee server
4. Bread toaster
5. Air cooler
6. Water heater

1. Advantages of open-loop control system

- Simple, easy to design, and economical due to less number of components
- Generally, open-loop systems are stable

2. Disadvantages of open-loop control system

- Inaccurate
- Unreliable
- The effect of parameter variation and external noise is more

Note: Open-loop control systems do not require performance analysis.

DC Gain

Consider a transfer function having a 'p' number of poles at the origin

$$\frac{K(s+s_1)(s+s_2)\ldots(s+s_n)}{s^p(s+s_a)(s+s_b)\ldots(s+s_m)}$$

or

$$\frac{K.s_1.s_2\ldots s_n\left(\frac{s}{s_1}+1\right)\left(\frac{s}{s_1}+1\right)\ldots\left(\frac{s}{s_n}+1\right)}{s^p.s_a.s_b\ldots s_m\left(\frac{s}{s_a}+1\right)\left(\frac{s}{s_b}+1\right)\ldots\left(\frac{s}{s_m}+1\right)}$$

(in time constant from)

Then its DC gain

$$= \lim_{s\to 0} s^p \frac{K.s_1.s_2\ldots s_n\left(\frac{s}{s_1}+1\right)\left(\frac{s}{s_1}+1\right)\ldots\left(\frac{s}{s_n}+1\right)}{s_p.s_a.s_b\ldots s_m\left(\frac{s}{s_a}+1\right)\left(\frac{s}{s_b}+1\right)\ldots\left(\frac{s}{s_m}+1\right)}$$

Example 1: The open-loop DC gain of a unity negative feedback system with a closed-loop transfer function $\frac{s+4}{(s^2+7s+13)}$ is

(A) $\frac{4}{13}$ (B) $\frac{4}{9}$

(C) 4 (D) 13

Solution: (B)

$$CLTF = \frac{G(s)}{1+G(s)H(s)} = \frac{s+4}{s^2+7s+13}$$

$$\frac{1+G(s)H(s)}{G(s)} = \frac{s^2+7s+13}{s+4}$$

H(s) = 1 for unity feedback system

$$\frac{1}{G(s)} = \frac{s^2+7s+13}{s+4} - 1 = \frac{s^2+6s+9}{s+4}$$

$$\therefore G(s) = \frac{s+4}{s^2+6s+9}$$

For DC gain $\omega = 0$ (or) s = 0

$$G(s) = open\ loop\ gain = \frac{4}{9}$$

There are two alternative forms of expressing a transfer function:

a) Pole-Zero Form

b) Time Constant Form

Closed-Loop Control System (C.L.C.S.)

If there is feedback from the output to the input terminal, then the system is termed as a Closed Loop Control System.

Since the feedback facilitates the system to automatically correct input to the system to meet desired response, it is also called an "Automatic control system."

Closed-loop control systems can be classified as positive and negative feedback control systems.

1. Transfer function of C.L.C.S. with negative feedback

The different signals in a closed loop control system are termed as:

R(s) = Reference input

C(s) = Output signal or controlled variable

B(s) = Feedback signal

E(s) = Actuating signal

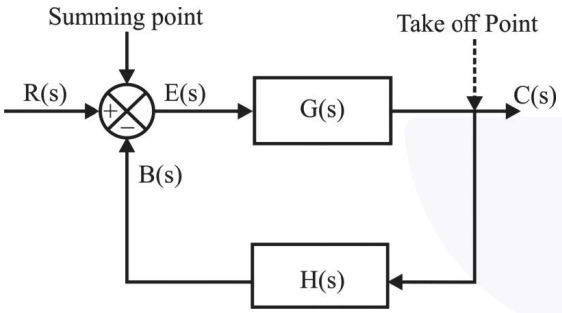


Fig. 2.4

$$G(s) = \frac{C(s)}{E(s)}$$ = Forward path transfer function

H(s) = Transfer function of feedback system or Feedback path transfer function

$$G(s)H(s) = \frac{B(s)}{E(s)}$$ = Loop transfer function

$$T(s) = \frac{C(s)}{R(s)}$$ = Closed – loop transfer function

$$C(s) = G(s).E(s)$$

E(s) = R(s) – B(s) = R(s) – H(s)C(s) ...(1)

$$E(s) = \frac{C(s)}{G(s)} \quad ...(2)$$

From (1) and (2)

$$\frac{C(s)}{G(s)} = R(s) - H(s)C(s) \Rightarrow R(s)$$

$$= C(s)\left[H(s) + \frac{1}{G(s)}\right]$$

$$\frac{C(s)}{R(s)} = T(s) = \frac{G(s)}{1 + G(s)H(s)}$$

- Closed-loop system can be represented by the following block diagram.

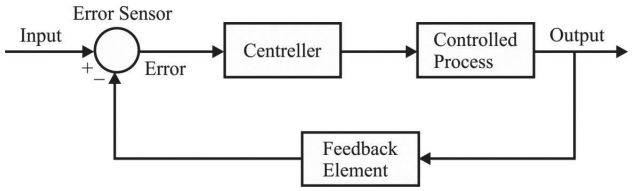


Fig. 2.5

Some common examples of closed-loop control systems are:

- Electric iron
- DC motor speed control
- A missile launching system (direction of missile changes with the location of moving target)
- Radar tracking system
- Human respiratory system
- Autopilot system
- Economic inflation
- Traffic signals operated based on the density of traffic
- Air conditioner

2. Advantages of closed-loop control systems

- Accurate and reliable
- Reduced effect of parameter variations
- Bandwidth of the system can be increased with negative feedback
- Reduced effect of non-linearities

3. Disadvantages of closed loop control systems

- The system is complex and costly
- Improper design of the controller may cause the system to become unstable
- Gain of the system reduces with negative feedback
- Feedback may lead to an oscillatory response

Points to Remember

- Feedback is not used to improving stability
- An open-loop stable system may also become unstable when negative feedback is applied

- Except for oscillators, in positive feedback, we always have unstable systems.

Difference between open-loop and closed-loopcontrol systems

Open-Loop System	Closed-Loop System
1. As long as the calibration is good, the open-loop system will be accurate	1. Due to feedback, the closed-loop system is more accurate
2. Organisation is simple and easy to construct	2. Complicated and difficult
3. Generally stable in operation	3. Stability depends on system components
4. If nonlinearity is present, system operation degrades	4. Comparatively, the performance is better than open-loop system if nonlinearity is present

Table 2.1

Example 2. Match List-I (Physical action or activity) with List-II (Category of the system) and select the correct code:

List-I

A. Human respiration system
B. Pointing at an object with a finger
C. A man driving a car
D. A thermostatically controlled room heater

List-II

I. Man-made control system
II. Natural including biological control system
II. Control system whose components are both man-made and natural

Codes:

	A	B	C	D
(A)	II.	II.	III.	I.
(B)	III.	I.	II.	I.
(C)	III.	II.	II.	III.
(D)	II.	I.	III.	III.

Solution: (A)

Example 3. If the open-loop transfer function of a negative feedback control system is $G(s)H(s)=\frac{K}{(s+1)^3}$, then the gain K for a closed-loop pole to be at $\left(-\frac{1}{2}+j\frac{\sqrt{3}}{2}\right)$ is

(A) 1 (B) 2
(C) 3 (D) 4

Solution: (A)

Open-loop transfer function,

$$G(s)=\frac{K}{(s+1)^3}$$

Closed-loop transfer function,

$$\frac{G(s)}{1+G(s)H(s)}=\frac{K}{(s+1)^3+K}$$

At $s=\left(-\frac{1}{2}+j\frac{\sqrt{3}}{2}\right)$

The denominator should be zero.

$$\therefore (s+1)^3+K=0$$

$$\left[-\frac{1}{2}+j\frac{\sqrt{3}}{2}+1\right]^3+K=0$$

$$\left[\frac{1}{2}+\frac{\sqrt{3}}{2}j\right]^3+K=0$$

$$1-j3\sqrt{3}+j3\sqrt{3}-9+8K=0$$

K = 1

Example 4. What is the unit step response of a continuous system whose transfer function has a zero at –1, a pole at –2, and a gain factor of 2?

Solution:

The Laplace transform of the output is given by Y(s) = G(s) U(s)

$$G(s)=\frac{2(s+1)}{(s+2)} \qquad U(s)=\frac{1}{s}$$

$$Y(s) = \frac{2(s+1)}{s(s+2)} = \frac{1}{s} + \frac{1}{s+2}$$

Evaluating the Inverse Laplace transform of the partial fraction expansion of Y(s), we have, $y(t) = 1 + e^{-2t}$.

Important Components of Block Diagram

- When a control system is represented as a connection of multiple blocks, then we need to simplify that block diagram in order to find the transfer function of the system.
- The following elements are generally present in the block diagram:

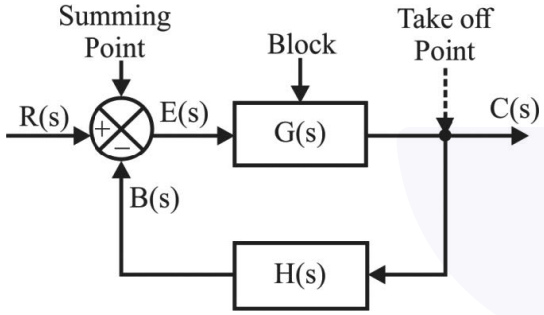
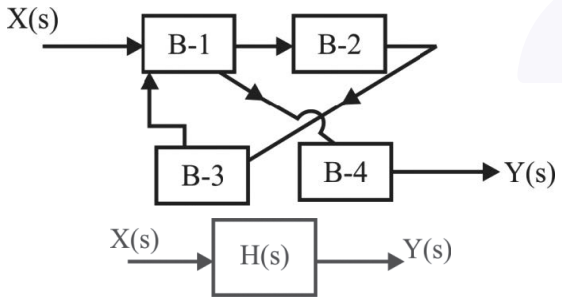


Fig. 2.6

Requirement: Block diagram reduction



Notations: Input → r(t) or R(s)
Output → c(t) or c(s)
Transfer function → G(s)

Simplification:

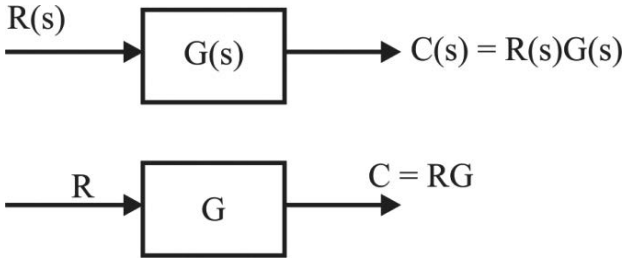
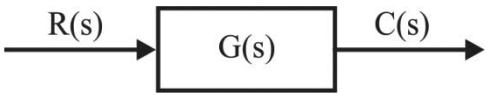


Fig. 2.7

1. Block

It is used to represent the transfer function of a component or sub-system.

The signal coming into the block represents the input signal [R(s)}, the signal going out of the block represents the output signal [C(s)} and the block itself represents the transfer function of a component.



Forward gain of the block $\frac{C(s)}{R(s)} = G(s)$

Fig. 2.8

2. Summer or Summing point

It is used to add/subtract two or more input signals.

Note: The signals that are added or subtracted must be of the same units.

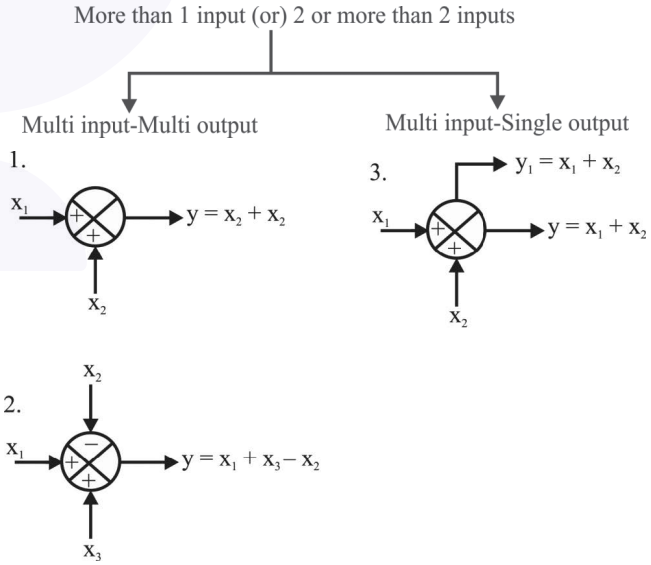
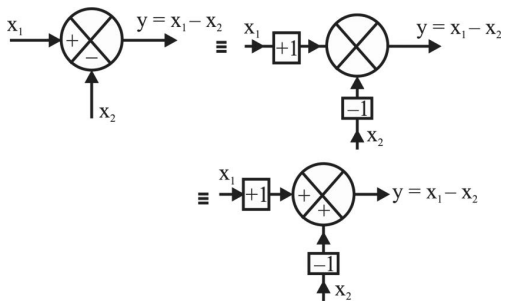


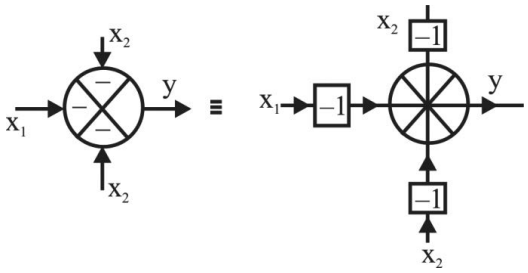
Fig.2.9

2. Making a summer neutral

a)



b)



Note: when summer is neutral, it can be represented by a "NODE" (point)

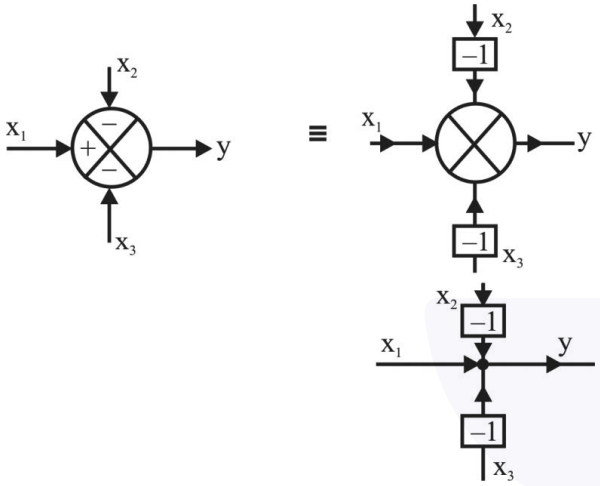
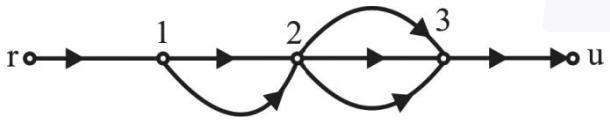
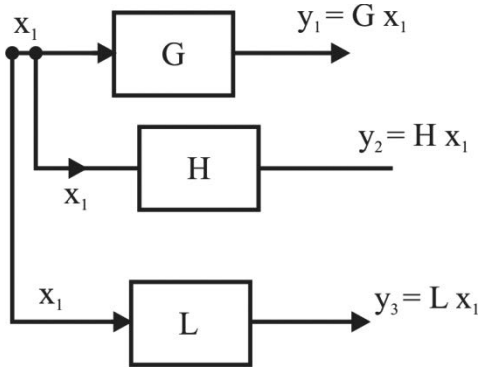


Fig. 2.10

4. **Take-off point (or) banch point:**"Any point is considered as a take-off point if it has a single input and multi-output".



1. Take off node
2. Summing node
3. Summing node

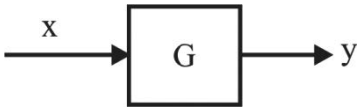


Note: "when multiple blocks need common input, then take-off point is used."

Node	Input	Output
Summing	More than 1 or ≥ 2	Multiple
Take off	1	Multiple

5. **Forward gain**

(a) Direction: From input towards output

(b) 

$$\frac{y}{x} = G$$

Block Diagram Reduction Rules

1. **Shifting of summing point after a block having gain 'G'**

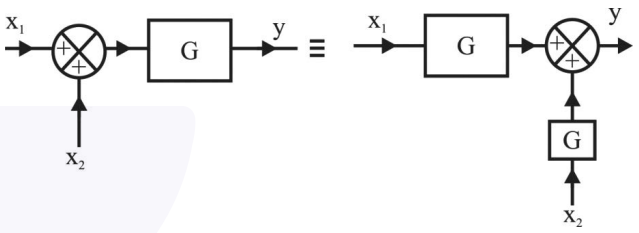


Fig. 2.11

2. **Shifting of summing point before a block having gain 'G'**

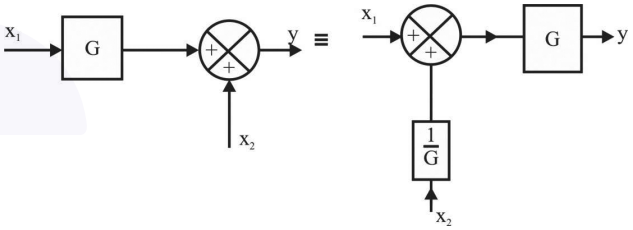


Fig. 2.12

3. **Shifting of take-off point after a block, having gain 'G'**

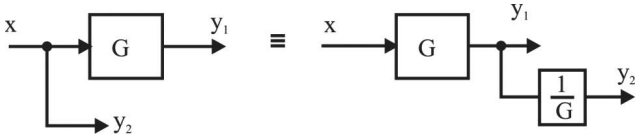


Fig. 2.13

4. **Shifting of take-off point before a block having gain 'G'**

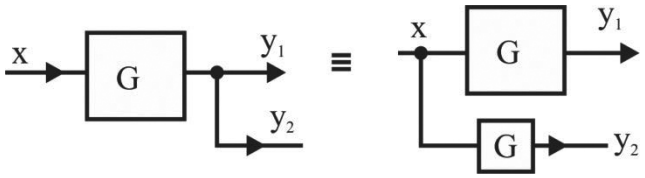


Fig. 2.14

5. Shifting of take-off point after summing point

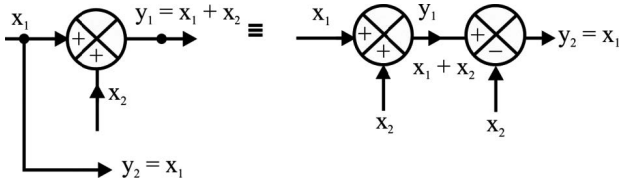


Fig. 2.15

Note: "In the above case,the second input of new summer will have an opposite sign".

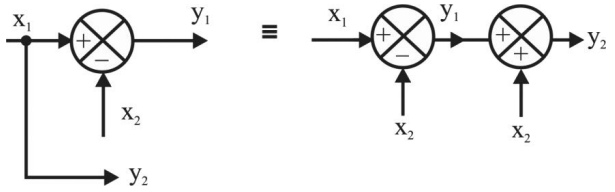


Fig. 2.16

6. Shifting of take-off point before summing point

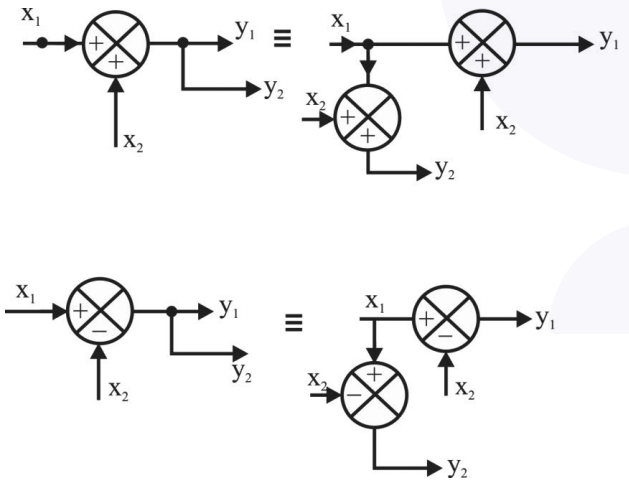


Fig. 2.17

Note: In the above case, the second input of new summer will have the same sign

7. Blocks or gains connected in parallel

two gains or blocks are connected in parallel

- Summing point should be common
- Input should be common

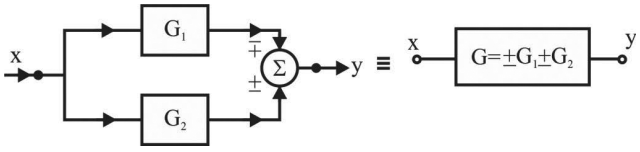


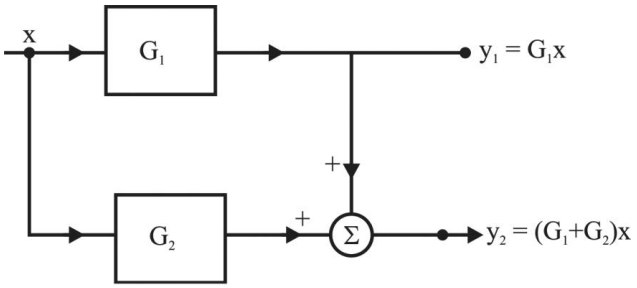
Fig. 2.18

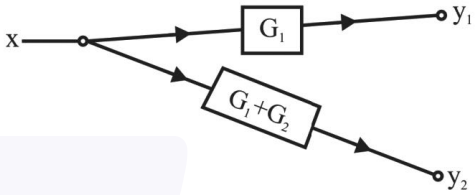
Resolve S-1 Hold nodes X and Y

S-2 Resultant will be present between X and Y:

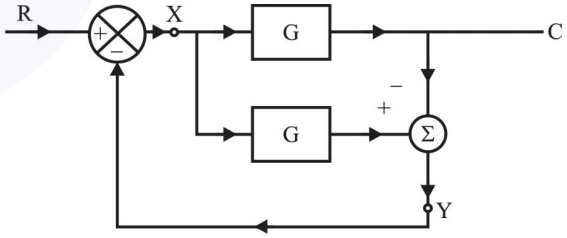
Examples

1.

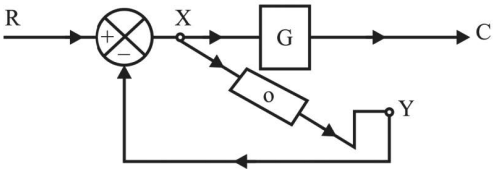




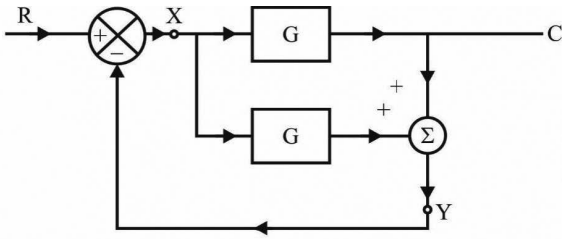
2.



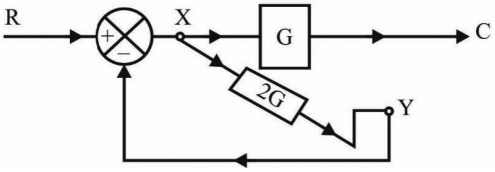
≡



3.



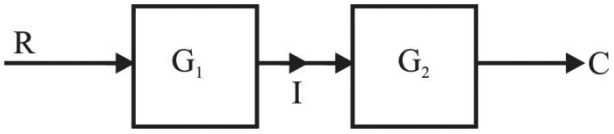
≡



8. Blocks or gains connected in cascade or series

If the output of the first block or component is directly given as input to the next block or component, then those two blocks are said to be connected in cascade.

Note: There should not be any summing point or take-off point for the blocks to be connected in cascade.



$$\frac{C}{R} = \begin{cases} G_1G_2 & I = 0 \text{ (no loading effect)} \\ G_1G_2 & I \neq 0 \text{ (loadingeffect)} \end{cases}$$

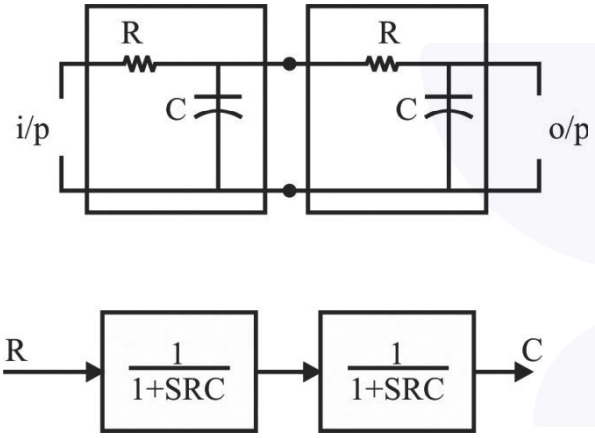
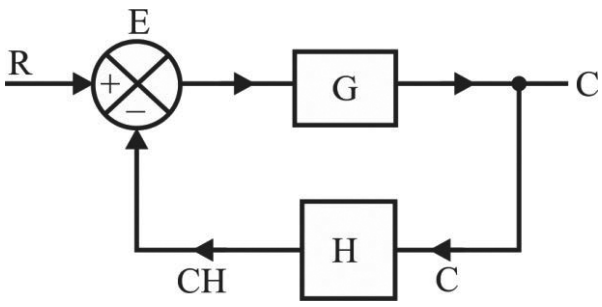


Fig. 2.19

Note: "In control systems, loading effect is ignored".

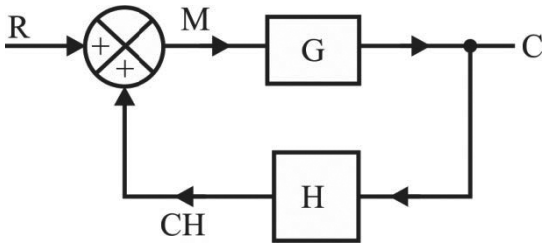
9. Elimination of feedback loop



E = R–CH (i)

C = EG = (R–CH) G C = RG–CHG

$$\frac{C}{R} = \frac{G}{1+GH}$$



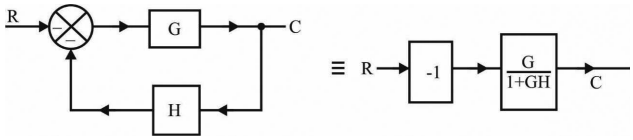
C(1 – GH) = RG

$$\frac{C}{R} = \frac{G}{1-GH}$$

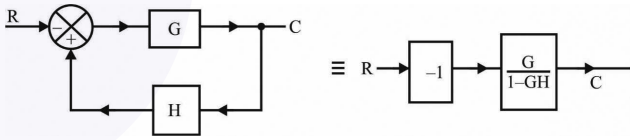
M = R + CH (i)

C = GM = G(R + CH)

C = RG + HGC



$$\frac{C}{R} = \frac{-G}{1+GH}$$



$$\frac{C}{R} = \frac{-G}{1-GH}$$

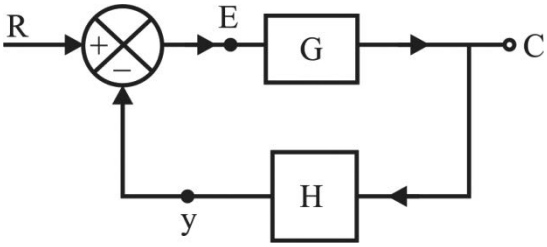
Note: "In any feedback circuit, only desired output needs to be shown"

Fig. 2.20

Common mistake:

Carefully observe the direction of arrows to differentiate between parallel and feedback blocks.

Example:

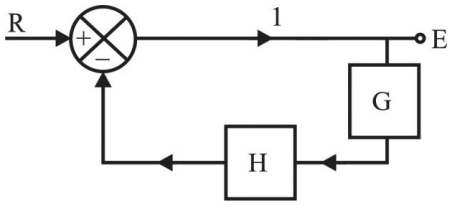


Calculate: (A) C/R (B) E/R (C) Y/R

Solution:

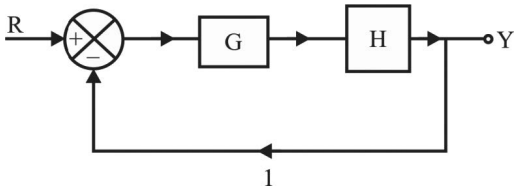
(A) $\frac{C}{R} = \frac{G}{1+GH}$

(B)



$$\frac{E}{R} = \frac{1}{1+GH}$$

(C)



$$\frac{Y}{R} = \frac{GH}{1+GH}$$

Problem-solving Technique

1. "Always try to eliminate common node by using two parallel paths".

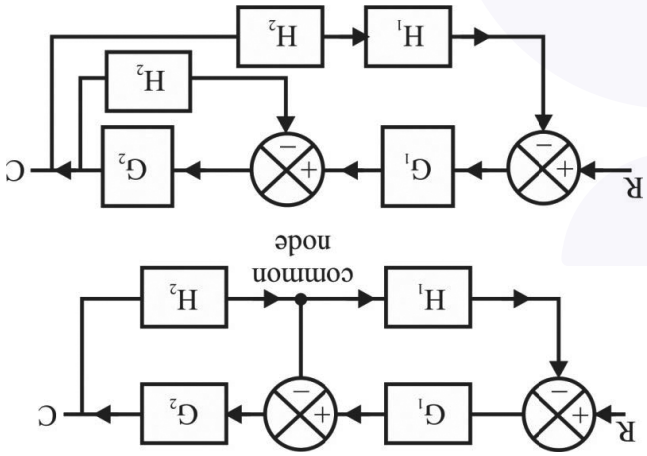


Fig. 2.21

2. Every 3 input summer must be converted into two 2 input summer:

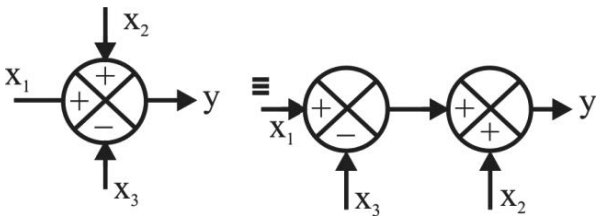


Fig. 2.22

3. Always try to bring two summer points side by side and change their respective inputs if needed.

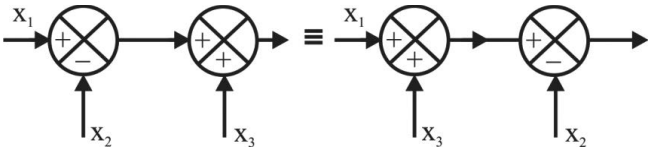
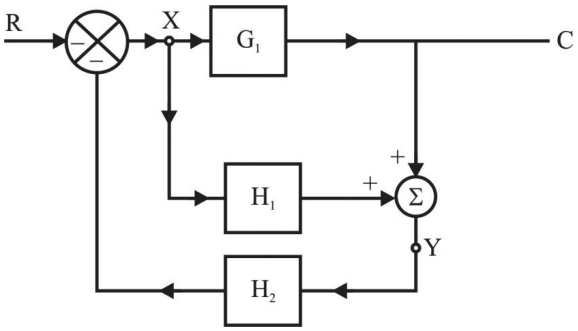


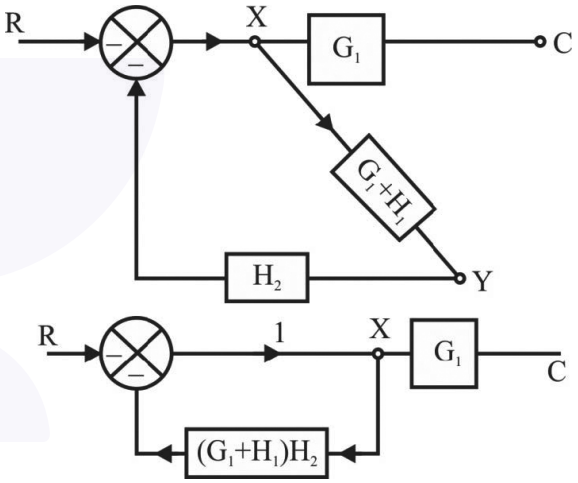
Fig. 2.23

Examples:



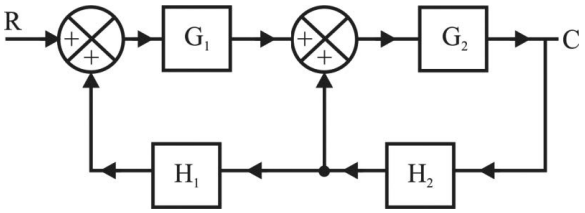
C/R =?

Solution:



$$\frac{C}{R} = \frac{(-1)\times G_1}{1+(G_1+H_1)H_2}$$

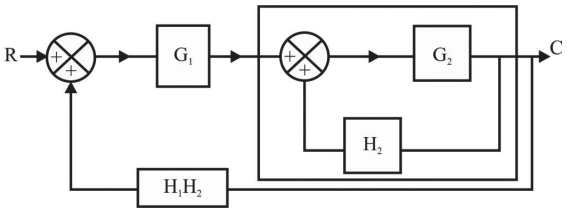
2.



Calculate C/R =?

Solution:

≡

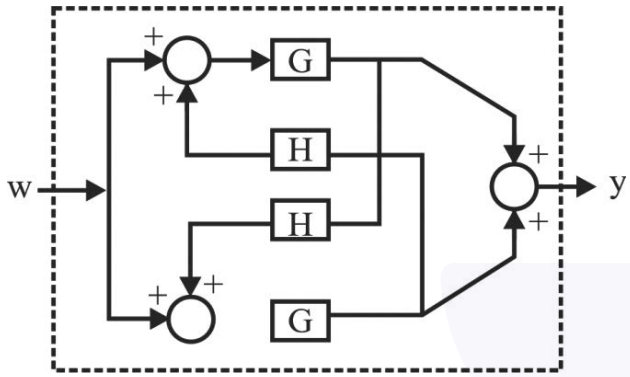


$$\frac{C}{R} = \frac{\dfrac{G_1G_2}{1-G_2H_2}}{1-\dfrac{G_1G_2H_1H_2}{1-G_2H_2}} = \frac{G_1G_2}{1-G_2H_2-G_1G_2H_1H_2}$$

Exercise Problems

1. The overall transfer function of the system shown in the figure is

[GATE 1992: 1 Mark]



(A) $\dfrac{G}{1-GH}$

(B) $\dfrac{2G}{1-GH}$

(C) $\dfrac{GH}{1-GH}$

(D) $\dfrac{GH}{1-H}$

Solution: (B)

2. The closed-loop transfer function of a control system is given by

$$\frac{C(s)}{R(s)} = \frac{2(s-1)}{(s+2)(s+1)}$$

For a unit step input, the output is

[GATE 1999: 1 Mark]

(A) $-3e^{-2t} + 4e^{-t} - 1$

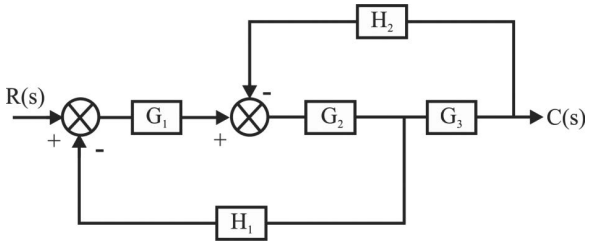
(B) $-3e^{-2t} - 4e^{-t} + 1$

(C) Zero

(D) infinity

Solution: (A)

3. For the block diagram shown in the figure, C(s)/R(s) is given by



[GATE 1998: 2 Marks]

(A) $\dfrac{G_1G_2G_3}{1+H_2G_2G_3+H_1G_1G_2}$

(B) $\dfrac{G_1G_2G_3}{1+G_1G_2H_1H_2}$

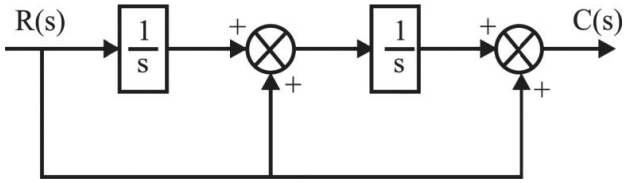
(C) $\dfrac{G_1G_2G_3}{1+G_1G_2G_3H_1+G_1G_2G_3H_2}$

(D) $\dfrac{G_1G_2G_3}{1+G_1G_2G_3H_1}$

Solution: (A)

4. For the block diagram shown in the figure, the transfer function $\dfrac{C(s)}{R(s)}$ is equal to

[GATE 2004: 2 Marks]



(A) $\dfrac{s^2+1}{s^2}$

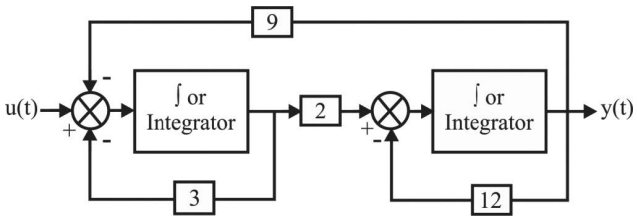
(B) $\dfrac{s^2+s+1}{s^2}$

(C) $\dfrac{s^2+s+1}{s}$

(D) $\dfrac{1}{s^2+s+1}$

Solution: (B)

5. The block diagram of a control system is shown in the figure. The transfer function G(s) = Y(s)/U(s) of the system is



[GATE 2003: 2 Marks]

(A) $\dfrac{1}{18\left(1+\dfrac{s}{12}\right)\left(1+\dfrac{s}{3}\right)}$

(B) $\dfrac{1}{27\left(1+\dfrac{s}{6}\right)\left(1+\dfrac{s}{9}\right)}$

(C) $\dfrac{1}{27\left(1+\dfrac{s}{12}\right)\left(1+\dfrac{s}{9}\right)}$

(D) $\dfrac{1}{27\left(1+\dfrac{s}{9}\right)\left(1+\dfrac{s}{3}\right)}$

Solution: (B)

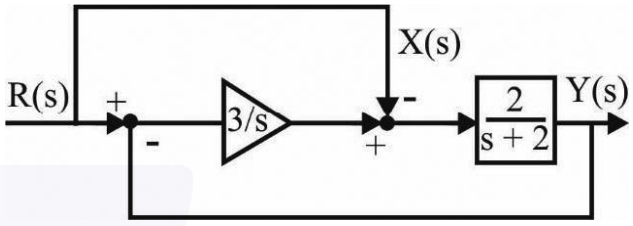
6. For a tachometer, if θ (t) is the rotor displacement in radians, e(t) is the output voltage, and K_t is the tachometer constant in V/rad/sec, then the transfer function $\dfrac{E(s)}{Q(s)}$ will be **[GATE 2004: 1 Mark]**

(A) $K_t s^2$ (B) $\dfrac{K_t}{s}$

(C) $K_t s$ (D) K_t

Solution: (C)

7. When subjected to a unit step input, the closed-loop control system shown in the figure will have a steady-state error of



[GATE 2005: 2 Marks]

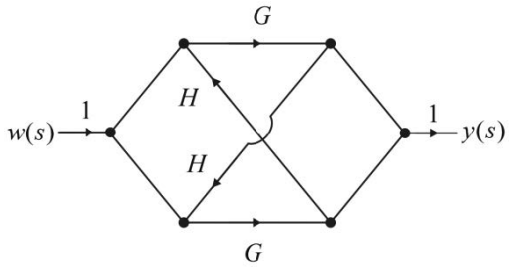
(A) -1.0 (B) -0.5

(C) 0 (D) 0.5

Solution: (C)

Exercise Problems

1. (B) SFG:



$$\therefore T.F. = \frac{G + GHG + G + GHG}{1 - [GHGH]}$$

[4 forward paths are there]

$$= \frac{2G}{1 - GH}$$

2. (A)

$$\frac{C(s)}{1/s} = \frac{2(s-1)}{(s+2)(s+1)}$$

$$C(s) = \frac{-1}{s} + \frac{-3}{s+2} + \frac{4}{s+1}$$

$$\therefore C(t) = -1 - 3e^{-2t} + 4e^{-t}$$

3. (A)

Number of forward paths

$= 1(G_1G_2G_3)$

Number of loops

$= 2(-G_1G_2H_1, -G_1G_3H_2)$

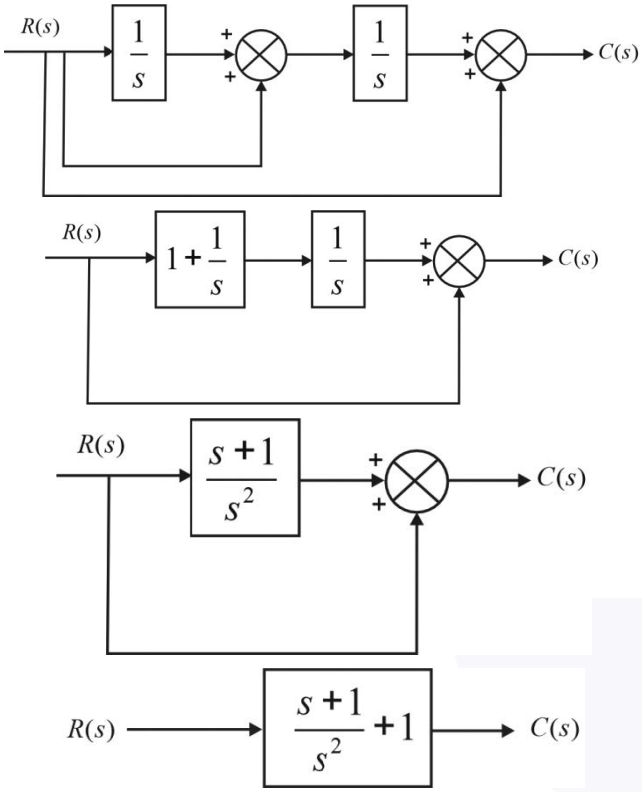
Number of non-touching loops = 0

$$\therefore T.F. = \frac{C(s)}{R(s)}$$

$$= \frac{G_1G_2G_3}{1 - G_1G_2H_1 + G_2G_3H_2}$$

4. (B)

Method -1: Using block-diagram reduction technique.

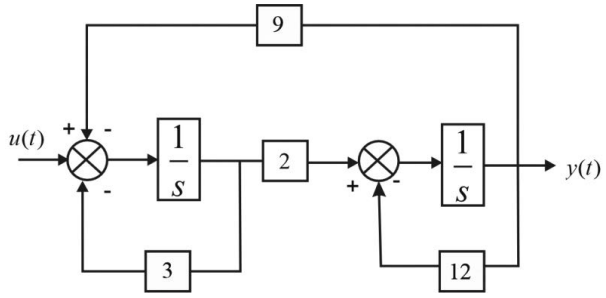


So, transfer function

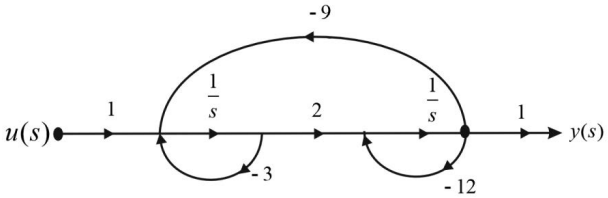
$$= \frac{C(s)}{R(s)} = \frac{s^2 + s + 1}{s^2}$$

5. (B)

Integrator are represented as 1/s in s-domain



As per the block diagram, the corresponding signal flow graph is drawn



One forward path $P_1 = 2/s^2$

The individual loop is,

$$L_1 = -\frac{3}{s}, L_2 = -\frac{12}{s} \text{ and } L_3 = -\frac{18}{s^2}$$

L_1 and L_2 are non-touching loos

$$L_1 L_2 = \frac{36}{s^2}$$

The loops touches the forward path $\Delta_1 = 1$

The graph determinant is

$$\Delta_1 = 1 - (L_1 + L_2 + L_3) + L_1 L_2$$

$$= 1 + \frac{3}{s} + \frac{12}{s} + \frac{18}{s^2} + \frac{36}{s^2}$$

Applying mason's gain formula:

$$G(s) = \frac{Y(s)}{U(s)} = \frac{P_1 \Delta_1}{\Delta}$$

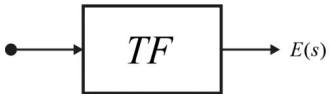
$$= \frac{2/s^2}{1 + \frac{3}{s} + \frac{12}{s} + \frac{18}{s^2} + \frac{36}{s^2}}$$

$$= \frac{2}{s^2 + 15s + 54}$$

$$= \frac{2}{(s+9)(s+6)}$$

$$= \frac{1}{27\left(1 + \frac{s}{9}\right)\left(1 + \frac{s}{6}\right)}$$

6. (C)



$\theta(t)$ = rotor displacement in radians

$\omega(t) = \frac{d\theta}{dt}$ = angular speedin rad/sec

Output voltage; $e(t) = K_t \omega(t) = K_t \frac{d\theta}{dt}$.

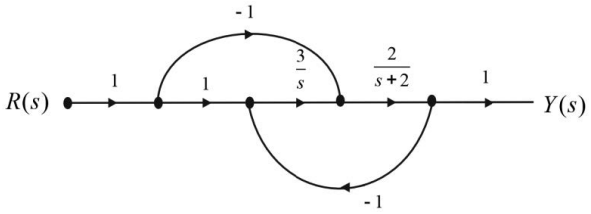
Taking Laplace to transform on both sides

$$E(s) = K_1 s\theta(s)$$

$$\Rightarrow \frac{E(s)}{\theta(s)} K_1 s$$

7. (C)

Using signal flow graph



Forward path gains

$$P_1 = (-1) \times \frac{2}{s+2} = \frac{-2}{s+2}$$

And $P_2 = \frac{3}{s} . \frac{2}{s+2} = \frac{6}{s(s+2)}$

Individual loop

$$L_1 = -1 \times \frac{3}{s} \times \frac{2}{s+2} = \frac{-6}{s(s+2)}$$

Loop touches forward paths, therefore,

$\Delta_1 = 1 \text{and} \Delta_2 = 1$

$D = 1 - L_1$

$$= 1 + \frac{6}{s(s+2)} = \frac{s(s+2)+6}{s(s+2)}$$

Using Mason's gain formula:

$$\frac{Y(s)}{R(s)} = \frac{P_1\Delta_1 + P_2\Delta_2}{\Delta}$$

$$= \frac{-\frac{2}{s+2} \times 1 + \frac{6}{s(s+2)} \times 1}{\frac{s(s+2)+6}{s(s+2)}}$$

$$\frac{Y(s)}{R(s)} = \frac{6-2s}{s^2+2s+6}$$

For unit step input, R(s) = 1/s

$$Y(s) = R(s).\left(\frac{6-2s}{s^2+2s+6}\right)$$

Error $= E(s) = R(s) - Y(s)$

$$= R(s) - R(s)\left(\frac{6-2s}{s^2+2s+6}\right)$$

$$= R(s)\left[1 - \frac{6-2s}{s^2+2s+6}\right]$$

$$E(s) = \frac{1}{s}\left[\frac{s^2+4s}{s^2+2s+6}\right]$$

Steady-state value of error, using final value theorem:

$$e_{ss} = \underset{s\to 0}{Lt}\, sE(s)$$

$$= \underset{s\to 0}{Lt}\, s\left[\frac{1}{s} . \frac{s^2+4s}{s^2+2s+6}\right]$$

$$= \underset{s\to 0}{Lt}\left[\frac{s^2+4s}{s^2+2s+6}\right] = 0$$

3 Signal Flow Graph

Objective

After completing this chapter user will be able to understand

- What is a signal flow graph?
- SFG reduction techniques
- Mason's gain formula
- Block diagram to SFG conversion
- How to find transfer function for complex block diagrams with the help of SFG

Introduction

- A Signal Flow Graph (SFG) is a graphical representation of the relationship between the variables of a set of linear algebraic equations.
- It graphically displays the transmission of signals through the system, as does the block diagram. But it is easier to draw and therefore, easier to manipulate than block diagrams.
- Besides the differences in the physical appearance of the SFG and the block diagram, the signal flow graph is constrained by rigid mathematical rules, whereas the block diagram notation is more liberal.

Signal flow graph:

- It is a graphical representation of a control system in which node is used to represent each system variable and are connected by direct branches.
- The branches are directed in the direction of signal flow, and the gain of each branch is the same as the transfer function of the block connected between those nodes. An example is shown below.

Block diagram:

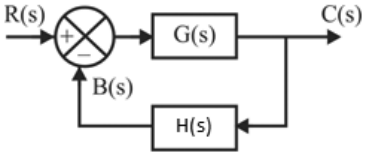


Fig. 3.1

Signal flow graph:

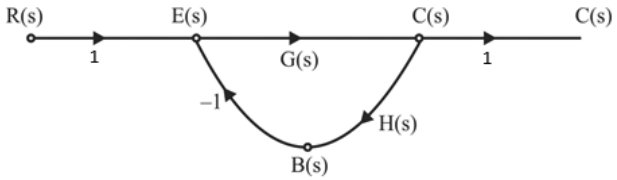
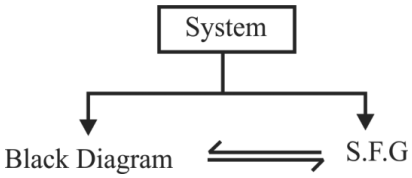


Fig. 3.2 Alternate Representation of a System.



Basic Properties of SFG

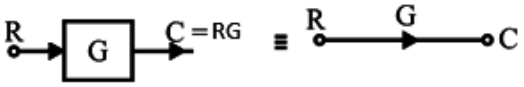
The important properties of the SFG are as follows:

1. SFG's are applicable only to linear systems.
2. The equations for which an SFG is drawn must be algebraic equations in the form of a cause and effect relationship.
3. Nodes are used to represent variables. Normally, the nodes are arranged from left to right, from input to output, following a succession of cause and effect relations through the system.
4. Signals travel along branches only in the direction described by the arrows of the branches.
5. The branch directing from nodes X_1 and X_2 represents the dependence of X_2 upon X_1, but not the reverse.

Correlation between Block Diagram and SFG

1. **Block:**

   A block with a transfer function can be represented as a branch with gain.



Lamped parameter Distributed parameter

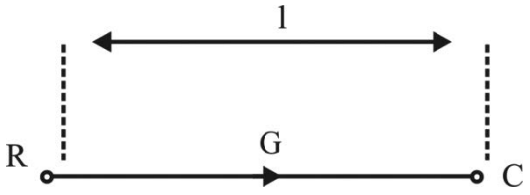


Fig. 3.3

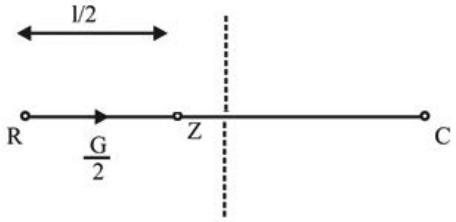


Fig. 3.4

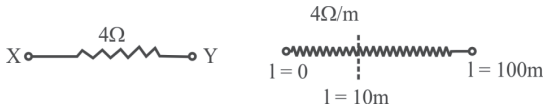


Fig. 3.5

$R = 4\Omega$

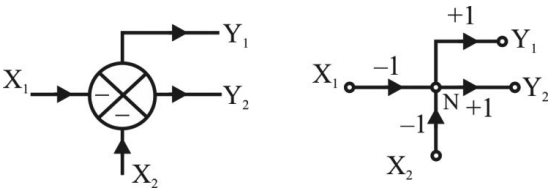
Lumped parameter $R = 40\Omega$

(Distributed parameter)

2. Summing point:

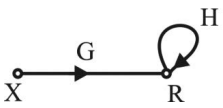
Summing point is represented by a node after making it as neutral.

Note: The value of a node is affected by only its incoming branches, and it is unaffected by outgoing branches.

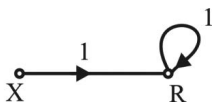


$N = -X_1 - X_2$ $y_1 = N \times 1$

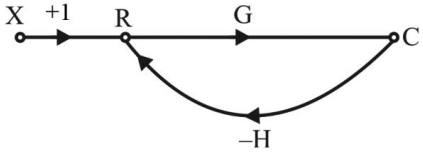
$y_2 = N \times 1$



$R = XG + RH$



$R = X + R$ (Inconsistent equation)



R = X − CH

3. Take–off point: It is also represented by a node.

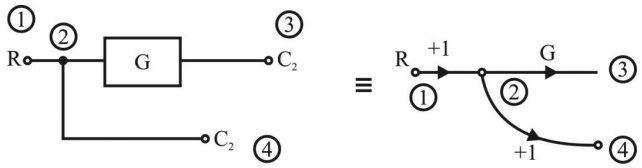


Fig. 3.6

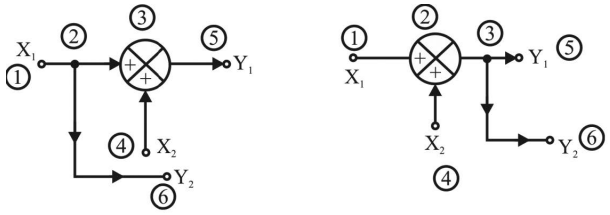


Fig. 3.7

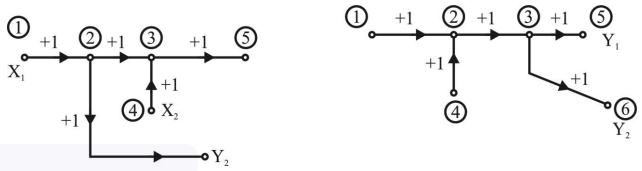
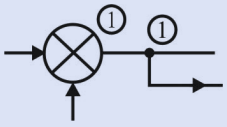
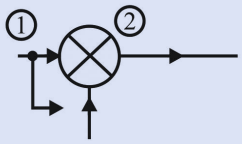


Fig. 3.8

Note: If the take-off point comes after summer, both take-off point and summer can be represented by the same node.

Summer	Single Node
Take–off point	Single node
	Single no node
	Two different nodes

Important Definitions

Input node vs initial node:

Input node or source:"It is the node having only outgoing branches."

Initial node: It may have both incoming and outgoing branches.

1.



R = X

(Input node)

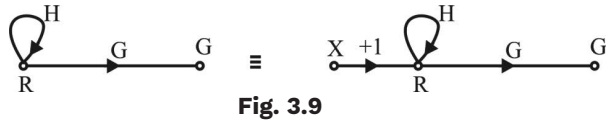
2. 

Fig. 3.9

R = X + RH, $R \neq X$

Not input node (Input node is 'X')

Initial node

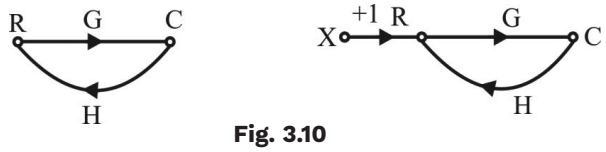
3. 

Fig. 3.10

R = X + CH

Initial node

Not input node

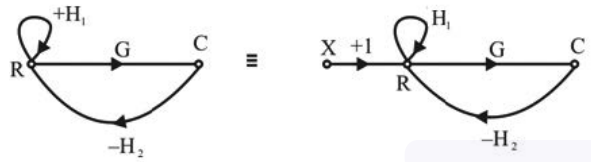
4. 

Fig. 3.11

$R = X + RH_1 - CH_2$

Initial node

Not input node

Output node or sink: This node has only incoming branches, but this condition may not be present always: so forward drag will full fill the above condition.

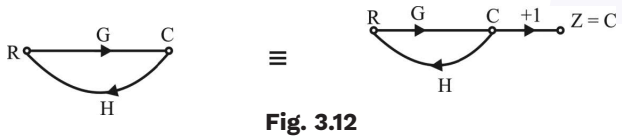


Fig. 3.12

Z = C

Note:

- Backward dragging may or may not result in the same potential or same node value.
- Forward dragging always results in the same potential or same node value.

Forward path:

- It is the path connecting the input node to the output node.
- Direction is always from input towards the output.
- Any node should not be traversed twice.
- Forward path should never be in loop form.

Loop:

- It is the path which originates and ends at the same node.
- Except for the origination node, all the other nodes must be traversed once.

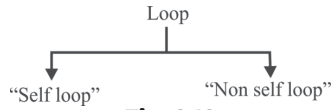


Fig. 3.13

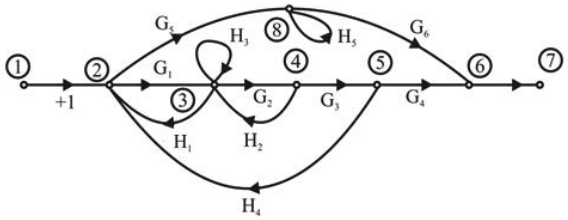


Fig. 3.14

Forward path	Forward path gain
$P_1 \rightarrow 1-2-3-4-5-6-7$	$G_1G_2G_3G_4$
$P_2 \rightarrow 1-2-8-6-7$	G_5G_6

Loop	Loop gain
$L_1 : 3-3$	H_3
$L_2 : 8-8$	H_5
$L_3 : 2-3-2$ *or* $3-2-3$	G_1H_1
$L_4 : 3-4-3$ *or* $4-3-4$	G_2H_2
$L_5 : 2-3-4-5-2$	$G_1G_2G_3H_4$

Two non-touching loops	Product of Loop gains
L_2L_1	H_3H_5
L_2L_3	?
L_2L_4	?
L_2L_5	?

Mason's Gain Formula

The relationship between an input variable and an output variable of a signal flow graph is given by the net gain between the input and output nodes and is known as the overall gain of the system.

Mason's gain formula is used to determine the overall gain of the system between input and output nodes.

Limitation:

It is used to calculate the ratio of:

- Output node to input node
- Intermediate node to input node

Note: Except for input and output, all other nodes are intermediate nodes.

Y - Output or Intermediate node

X - Input node

$$\frac{Y}{X} = \frac{\sum_{k=1}^{n} P_k \Delta_k}{\Delta} = \frac{P_1\Delta_1 + P_2\Delta_2}{\Delta} \quad \text{If n = 2}$$

$P_k \rightarrow$ Gain of Kth forward path

n $\rightarrow$Number of forward paths from X to Y.

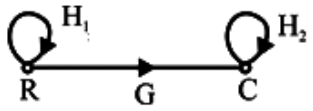
$\Delta \rightarrow$ It is independent of the forward path.

It considers all the loops present in SFG.

Δ = 1 – (sum of all loop gains) + sum of product of 2 non touching loop gains – sum of product of 3 non touching loop gains + ...

- $\Delta_k \rightarrow$ It is dependent on the forward path:
- $\Delta_k \rightarrow$ I – (sum of all loop gains not touching k^{th} forward path) + (sum of the product of 2 non-touching loop gain not touching k^{th} forward path) – (sum of the product of 3 non-touching loop gains not touching k^{th} forward path)

Example 1:



$\frac{C}{R} = ?$

Solution:

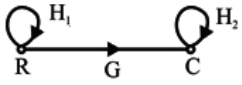
Method 1:

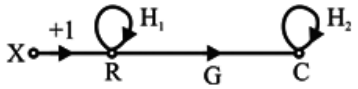
$C = RG + CH_2$

$C(1 - H_2) = RG$

$\frac{C}{R} = \frac{G}{1 - H_2}$

Method 2:





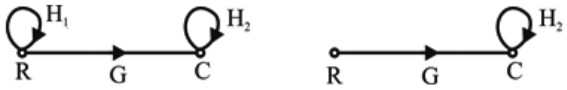
$\frac{C}{R} = ?$

$$\frac{R}{X} = \frac{P_1\Delta_1}{\Delta} = \frac{1(1 - H_2)}{\Delta} \quad \text{...(i)}$$

$$\frac{C}{X} = \frac{P_1\Delta_1}{\Delta} = \frac{G(1 - 0)}{\Delta} \quad \text{...(ii)}$$

$\frac{(i)}{(ii)}$:

Method 3:



$\frac{C}{R} = \frac{P_1\Delta_1}{\Delta}$

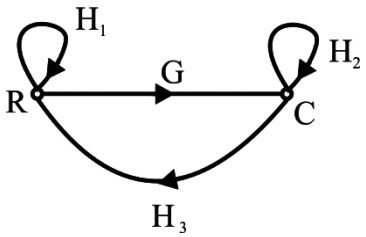
Note: Initial node + self-loop (Remove self-loop):

$\frac{C}{R} = \frac{P_1\Delta_1}{\Delta}$

$\frac{C}{R} = \frac{G(1 - 0)}{1 - H_2}$

$\frac{C}{R} = \frac{G}{1 - H_2}$

Example 2:



$\frac{C}{R} = ?$

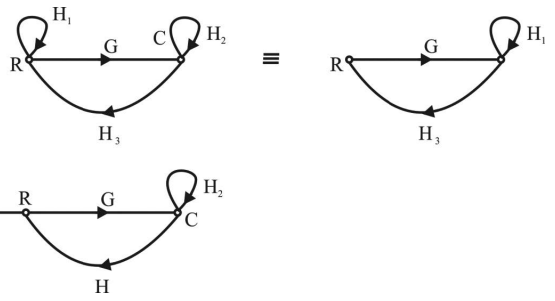
Solution:

Method 1:

$C = GR + CH_2$

$\frac{C}{R} = \frac{G}{1 - H_2}$

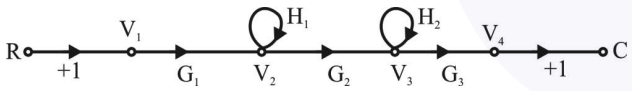
Method 2:



$$\frac{C}{R} = \frac{G}{1-H_2}$$

Note: "Self loop at initial node (of loop gain unity or non-unity) can always be ignored while applying Mason's gain formula: this may or may not make initial node as input node."

Example 3:



Solution:

Method 1: $V_1 = R$

$$V_2 = V_1G_1 + V_2H_1 \rightarrow V_2 = \frac{V_1G_1}{(1-H_2)} = \frac{RG_1}{(1-H_1)}$$

$$V_3 = V_2G_2 + V_3H_2 \rightarrow V_3 = \frac{V_2G_2}{(1-H_2)} = \frac{G_1G_2 \times R}{(1-H_1)(1-H_2)}$$

$$V_4 = V_3xG_3$$

$$\frac{C}{R} = \frac{G_1G_2G_3}{(1-H_1)(1-H_2)} \qquad \text{(i)}$$

Method 2:

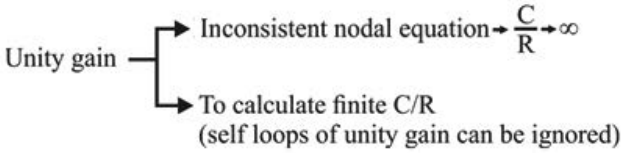
$$\frac{C}{R} = \frac{P_1\Delta_1}{\Delta} = \frac{G_1G_2G_3(1-0)}{1-(H_1+H_2)+H_1H_2} \qquad \text{(ii)}$$

(i) = (ii)

- $H_1 = 1 \quad \frac{C}{R} \rightarrow \infty$
- $H_2 = 1 \quad \frac{C}{R} \rightarrow \infty$
- $H_1 = 1 \;\; and \;\; H_2 = 1 \frac{C}{R} \rightarrow \infty$

Note:

- Self-loop at the initial node of unity or non-unity gain can be ignored, and the answer remains the same.
- Self-loops at intermediate nodes having



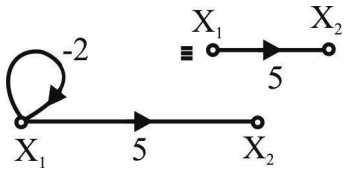
Points to Remember

- Carefully observe the forward path (arrow direction)
- Carefully observe the loops/non-touching loops
- Take care of loop at input node

Loop at Input Node

We are explaining how to handle the loop at the input node with a few examples.

Example 1: Find the value of $\frac{X_2}{X_1} = ?$



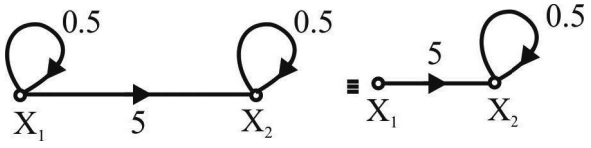
Removed the loop at X_1

Example 2:

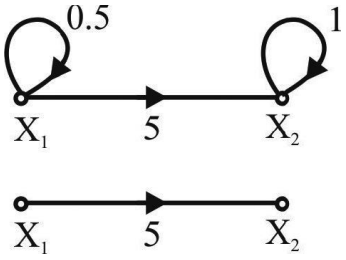
Find $\frac{X_2}{X_1} = ?$

$$X_2 = 5X_1 + 0.5X_2$$

$$\frac{X_2}{X_1} = 10$$



Example 3: Find the value of $\frac{X_2}{X_1} = ?$

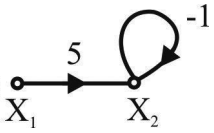


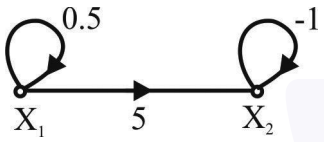
$\frac{X_2}{X_1} = 5$

Example 4: Find the value of $\frac{X_2}{X_1}$ =?

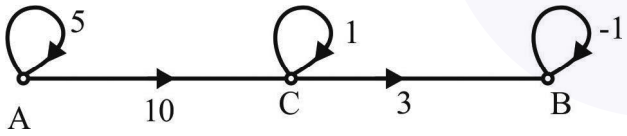
$X_2 = 5X_1 - X_2$

$\frac{X_2}{X_1} = 2.5$ ≡





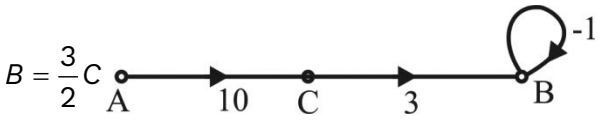
Example 5:



Calculate $\frac{B}{A} = ?$

Sol: C = 10A

B = 3C – B

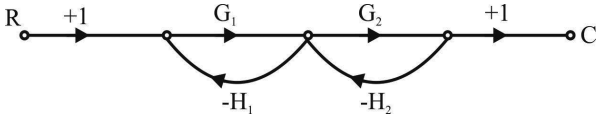


$B = \frac{3}{2} \times 10A$

$\frac{B}{A} = 15.$

Steps to find transfer function:

Example 6:



Calculate $\frac{C}{R} = ?$

Solution:

Step 1: $\frac{C}{R} \Rightarrow M.G.$

Step 2: Find out the forward path gain:

$P_1 = G_1G_2 \qquad \frac{C}{R} = \frac{P_1\Delta_1}{\Delta}$

Step 3:Calculation of loop gains:

1.Individual loop gains: $L_1 = -G_1H_1$

$L_2 = -G_2H_2$

2.Non-touchingloops = 0

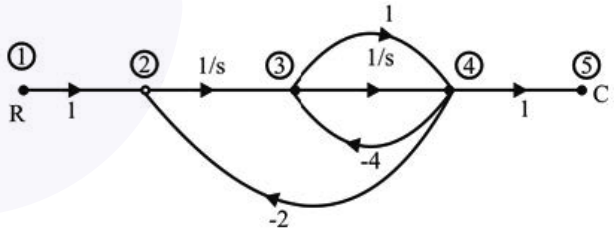
Step 4: Calculation of Δ_k *and* Δ

$\Delta = 1 - (L_1 + L_2) + (0)$

$\Delta_1 = 1 - (0)$

Step 5: $\frac{C}{R} = \frac{G_1G_2}{1-(-G_1H_1 - G_2H_2)} = \frac{G_1G_2}{1+G_1H_1 + G_2H_2}$

Example 7:



Calculate $\frac{C}{R} = ?$

Solution:

S.1 $\frac{C}{R} \rightarrow M.G.$

S.2 $P_1 = \frac{1}{s^2} \qquad \frac{C}{R} = \frac{P_1\Delta_1 + P_2\Delta_2}{\Delta}$

$p_2 = \frac{1}{s}$

S.3 *Individual Loops* $L_1 = \frac{-2}{s^2} \quad L_4$

$= \frac{-2}{s} \quad L_1 \rightarrow 2-3-4-2,\ L_2 \rightarrow 3-4-3$

$L_2 = -4/s \quad$ *Non Touching* $\rightarrow 0 \quad L_3 = 3-4-3$

$L_3 = -4 \qquad L_4 \rightarrow 2-3-4-2$

S.4: $\Delta = 1 - (L_1 + L_2 + L_3 + L_4)$

$$= 1 - \left\{\frac{-2}{s^2} - \frac{4}{s} - 4 - \frac{2}{s}\right\}$$

$$\Delta = \left\{1 + \frac{2}{s^2} + \frac{4}{s} + 4 + \frac{2}{s}\right\} = \left\{\frac{2}{s^2} + \frac{6}{s} + 5\right\}$$

$$\Delta_1 = 1 - (0)$$

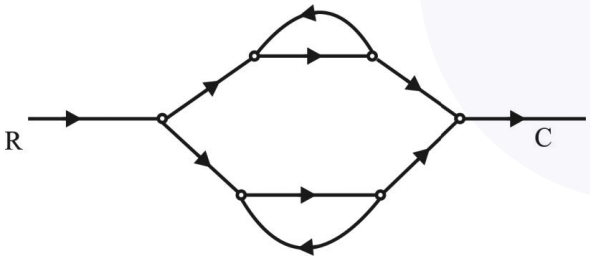
$$\Delta_2 = 1 - (0)$$

S.5: $\frac{C}{R} = \frac{\frac{1}{s^2} \times 1 + \frac{1}{s} \times 1}{\frac{2}{s^2} + \frac{6}{s} + 5} = \frac{1+s}{5s^2 + 6s + 2}$

$$\frac{C}{R} =$$

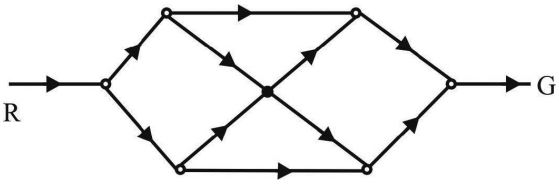
Few Important Cases

Case: 1



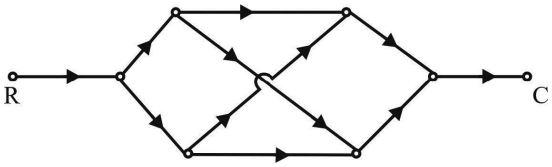
Forward paths - 2
Individual loops – 2
2 non–touching loops – 1

Case: 2



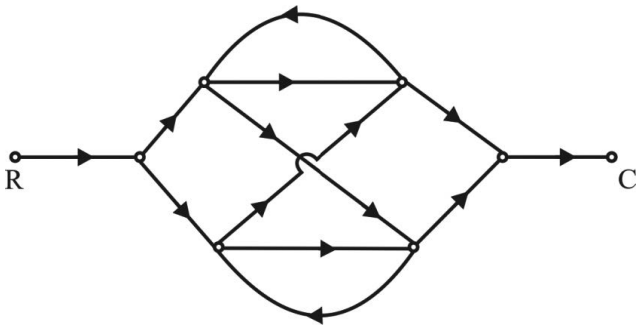
F.P.'s – 6
Loop's – 0
2 non-touching loops – 0

Case: 3



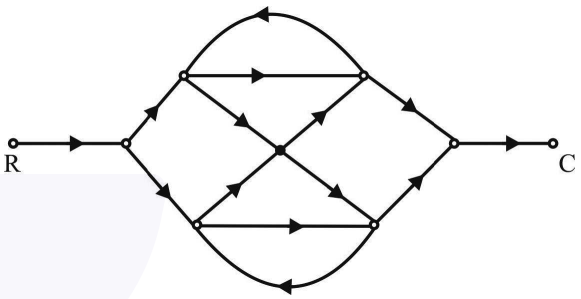
F.P.'s – 4
Loops – 0
2 non-touching loops– 0

Case: 4



F.P.'s– 6
Loops – 3
2 non–touching loops– 1

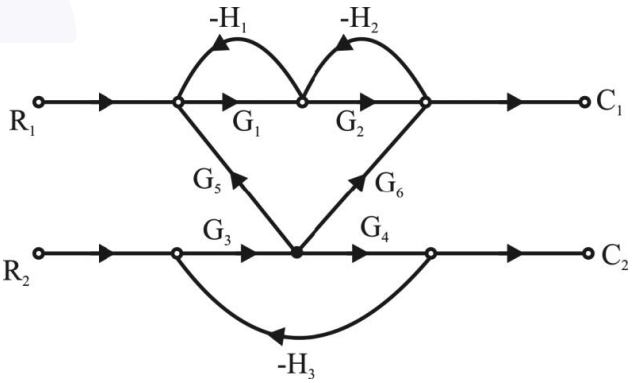
Case: 5



F.P.'s– 6
Loops – 4
2 non–touching loops– 3

Multi-input and multi-output SFG

Example 1:



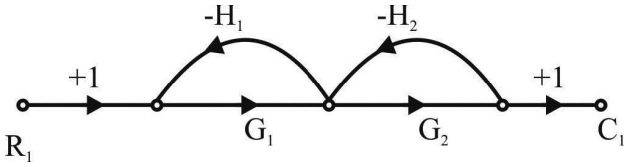
Determine

(A) $\left.\frac{C_1}{R_1}\right|_{R_2=0}$ (B) $\left.\frac{C_1}{R_2}\right|_{R_1=0}$

(D) $\left.\frac{C_2}{R_2}\right|_{R_1=0}$ (C) $\left.\frac{C_2}{R_1}\right|_{R_2=0}$

Solution:

(A) $\left.\frac{C_1}{R_1}\right|_{R_2=0}$



$$\left.\frac{C_1}{R_1}\right|_{R_2=0} = \frac{G_1G_2}{1+G_1H_1+G_2H_2}$$

(B) $\left.\frac{C_1}{R_2}\right|_{R_1=0}$ $P_1 = G_3G_6$

$P_2 = G_3G_5G_1G_2$

$$\left.\frac{C_1}{R_2}\right|_{R_1=0} = \frac{P_1\Delta_1 + P_2\Delta_2}{\Delta}$$

Individual Loops:

$L_1 = -G_1H_1$ $\quad 2N.T.$ $\quad L_1L_3$

$L_2 = -G_2H_2$ $\quad L_2L_3$

$L_3 = -G_3G_4H_3$

$\Delta = 1-(L_1+L_2+L_3)+(L_1L_3+L_2L_3)$

$\Delta_1 = 1+G_1H_1$

$\Delta_2 = 1-0$

$$\frac{C}{R} = \frac{P_1\Delta_1 + P_2\Delta_2}{\Delta}$$

(C) $\left.\frac{C_2}{R_2}\right|_{R_1=0}$

$P_1 = G_3G_4$

$\left.\frac{C_2}{R_2}\right|_{R_1=0} = \frac{P_1\Delta_1}{\Delta} \rightarrow calculate$

I.L.2 N.T.

L_1 $\quad L_1L_3$

L_2 $\quad L_2L_3$

L_3 $\quad L_2L_3$

$\Delta = same$

$\Delta_1 = 1+G_1H_1+G_2H_2$

(D) $\left.\frac{C_2}{R_1}\right|_{R_2=0} = 0$ (Since there are no forward paths)

Calculation of Transfer Function between Intermediate Nodes

Mason gain formula:

$R \rightarrow Input\ Node$

$C \rightarrow Output\ Node$

$y_1, y_2 \rightarrow Intermediate\ Node$

$$\frac{y_2}{y_1} = \frac{(y_2/R)}{(y_1/R)}$$

$$\frac{y_2}{y_1} = \frac{(y_2/R)}{(y_1/R)} = \left(\frac{\frac{\sum_{k=1}^{n} P_k\Delta_k}{\Delta}}{\frac{\sum_{l=1}^{m} P_l\Delta_l}{\Delta}}\right) = \frac{\sum_{k=1}^{n} P_k\Delta_k}{\sum_{l=1}^{m} P_l\Delta_l}$$

Note: While calculating ratio of intermediate nodes, Δ need not to be calculated.

Example:

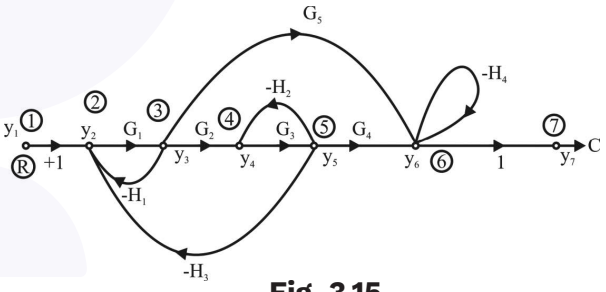


Fig. 3.15

(i) $\frac{y_7}{y_2}$

Solution:

(i) $\frac{y_7}{y_2} = \frac{(y_7/R)}{(y_2/R)}$

$$\frac{y_7}{R} = \frac{y_7}{y_1}$$

1. $P_1 = G_1G_2G_3G_4$ $\qquad \frac{y_7}{y_1} = \frac{P_1\Delta_1 + P_2\Delta_2}{\Delta}$

$P_2 = G_1G_5$

2. Loops:

$L_1 = -G_1H_1(2-3-2)$ $\quad 2N.T.$ $\quad 3N.T.$

$L_2 = -G_3H_2(4-5-4)$ $\quad L_1L_3$ $\quad L_1L_2L_3$

$L_3 = -H_4(6-6)$ L_3L_4

$L_4 = -G_1G_2G_3H_3$ L_1L_2

$(2-3-4-5-2)$ L_2L_3

3. $\Delta_1 = 1-0$

$\Delta_2 = 1+G_3H_2$

4. $\frac{y_7}{y_1} = \frac{P_1\Delta_1 + P_2\Delta_2}{\Delta} - (i)$

$\frac{y_2}{y_1}$

1. $P_1 = 1$ $\frac{P_1\Delta_1}{\Delta} = \frac{y_2}{y_1}$

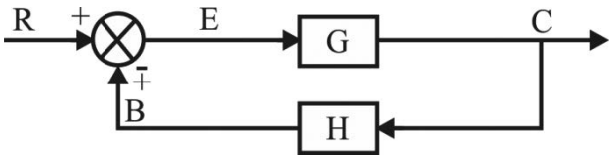
2. $\Delta_1 = 1-(L_2+L_3)+L_2L_3$

3. $\frac{y_2}{y_1} = \frac{P_1\Delta_1}{\Delta} - (ii)$ $\frac{y_7}{y_2} = \frac{(i)}{(ii)}$

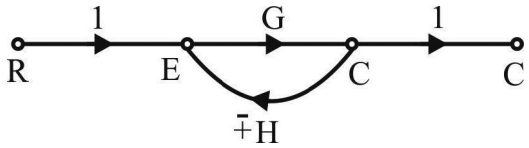
Drawing SFG From a Given Block Diagram

- The signal flow graph of a linear feedback control system whose components are specified by non-interacting transfer functions can be constructed by direct reference to the block diagram of the system.
- Each variable of the block becomes a node and each block having a transfer function becomes a branch with gain.
- While drawing a signal flow graph from a given block diagram, the adjacent summing points and take-off points (but not take–off pointspreceding a summing point in the direction of signal flow) are represented by a line joining the respective nodes. The direction of signal flow is indicated by an arrow on the line.
- However, in the direction of signal flow, if a take-off point precedes a summing point, then such points are represented by two separate nodes with a transmittance/gain of unity between them.

Example: The block diagram of the canonical feedback control system is given in the figure. Draw its corresponding SFG.

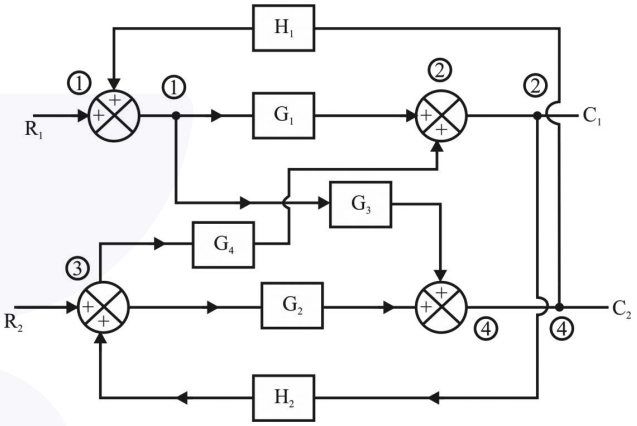


Solution: The signal flow graph is easily constructed from the figure. Note that the – or + sign of the summing point is associated with H.

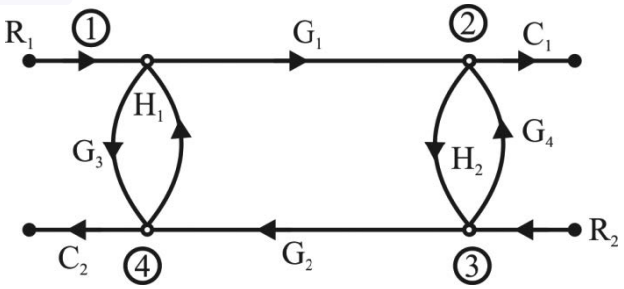


Example 2.

$\left.\frac{C_2}{R_1}\right|_{R_2=0} = ?$



Solution:



1. $P_1 = G_1G_2H_2$

$P_2 = G_3$

$\left.\frac{C_2}{R_1}\right|_{R_2=0} = \frac{P_1\Delta_1 + P_2\Delta_2}{\Delta}$

2. $L_1 = G_3H_1$ 2 N.T.

$L_2 = G_4H_2$ L_1L_2

$L_3 = G_1G_2H_1H_2$

3. $\Delta_1 = 1 - 0$

$\Delta_2 = (1 - G_4H_2)$ $\Delta = 1 - (L_1 + L_2 + L_3) + L_1L_2$

4. $\left.\frac{C_2}{R_1}\right|_{R_2=0} = \frac{P_1\Delta_1 + P_2\Delta_2}{\Delta}$

Test Your Knowledge

1. Which one of the four signal flow graphs shown in (A), (B), (C) and (D) represents the block diagram shown in the below figure?

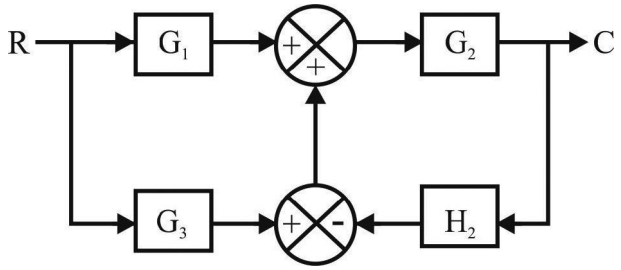
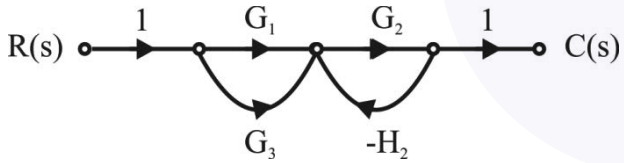
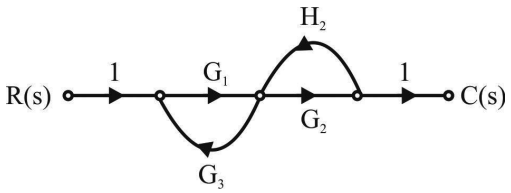


Fig. 3.16

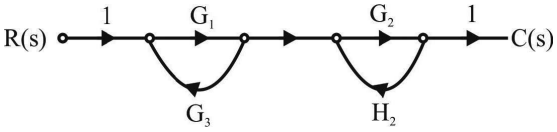
(A)



(B)



(C)



(D)

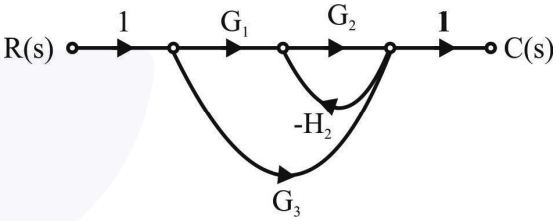


Fig. 3.17

Solution: (A)

Exercise Problems

1. The signal flow graph representation of a control system is shown below.

The transfer function $\frac{Y(s)}{R(s)}$ is computed as:

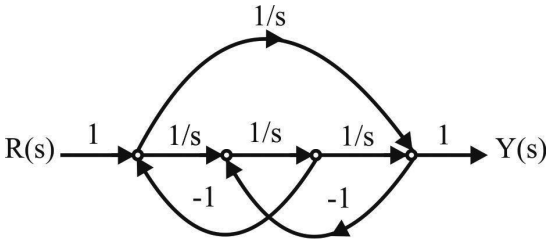


Fig. 3.18

(A) $\frac{1}{s}$

(B) $\frac{s^2+1}{s(s^2+2)}$

(C) $\frac{s(s^2+1)}{s^2+2}$

(D) $1 - \frac{1}{s}$

Solution: (A)

2. The transfer function between y_2 *and* y_1 in the figure is

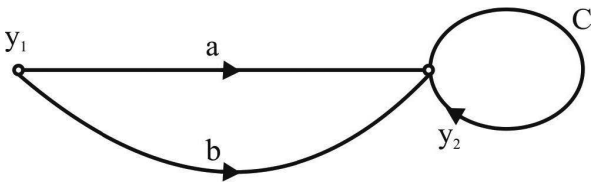


Fig. 3.19

(A) a + b

(B) (a + b)c

(C) $\frac{a+b}{1-c}$

(D) $\frac{a+b}{1+c}$

Solution: (C)

3. In the signal flow graph shown in figure if $X_2 = TX_1$ then T is equal to:

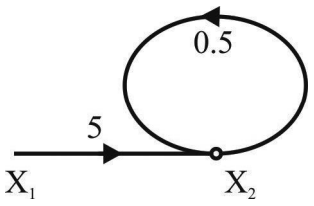
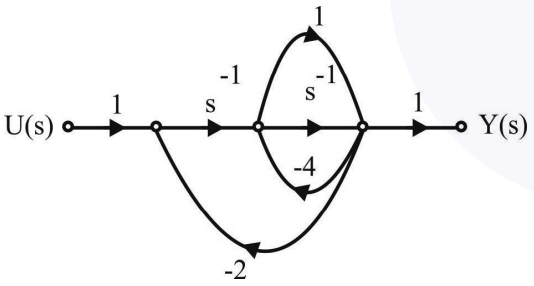


Fig. 3.20

(A) 2.5
(B) 5
(C) 5.5
(D) 10

Solution: (D)

4. The signal flow graph for a system is given below. The transfer function $\frac{Y(s)}{U(s)}$ for this system is: **[GATE 2013: 2 Marks]**



(A) $\frac{s+1}{5s^2+6s+2}$

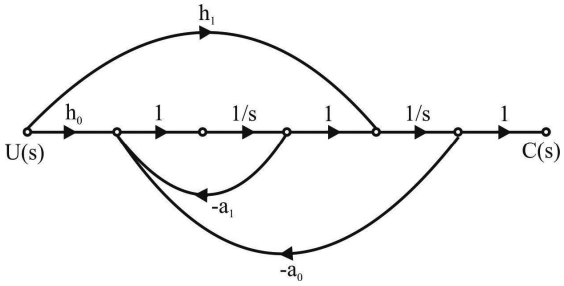
(B) $\frac{s+1}{s^2+6s+2}$

(C) $\frac{s+1}{s^2+4s+2}$

(D) $\frac{1}{5s^2+6s+2}$

Solution: (A)

5. The signal flow graph of a system is shown below. U(s) is the input, and C(s) is the output.



Assuming, $h_1 = b_1$ *and* $h_0 = b_0 - b_1a_1$.The input-output transfer function, $G(s) = \frac{C(s)}{U(s)}$ of the system is given by

[GATE 2014: 1 Marks, Set-3]

(A) $G(s) = \frac{b_0 s + b_1}{s^2 + a_0 s + a_1}$

(B) $G(s) = \frac{a_1 s + a_0}{s^2 + b_1 s + b_0}$

(C) $G(s) = \frac{b_1 s + b_0}{s^2 + a_1 s + a_0}$

(D) $G(s) = \frac{a_0 s + a_1}{s^2 + b_0 s + b_1}$

Solution: (C)

6. For the signal flow graph shown in the figure, which one of the following expressions is equal to the transfer function $\left.\frac{Y(s)}{X_2(s)}\right|_{X_1(s)=0}$?

[GATE 2015: 1 Marks, Set-1]

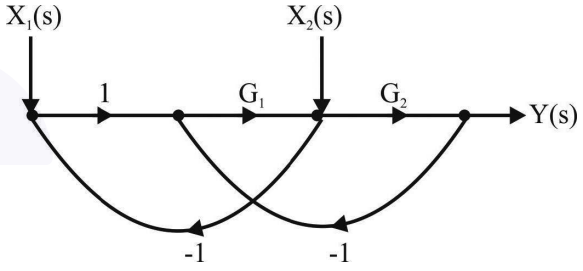


Fig. 3.21

(A) $\frac{G_1}{1+G_2(1+G_1)}$ (B) $\frac{G_2}{1+G_1(1+G_2)}$

(C) $\frac{G_1}{1+G_1G_2}$ (D) $\frac{G_2}{1+G_1G_2}$

Solution: (B)

7. For the system governed by the set of equations:

$$\frac{dx_1}{dt} = 2x_1 + x_2 + u$$

$$\frac{dx_2}{dt} = -2x_1 + u$$

$$y = 3x_1$$

The transfer function $\frac{Y(s)}{U(s)}$ is given by

[GATE 2015:2 Marks, Set-2]

(A) $\frac{3(s+1)}{(s^2-2s+2)}$

(B) $\frac{3(2s+1)}{(s^2-2s+1)}$

(C) $\frac{(s+1)}{(s^2-2s+1)}$

(D) $\frac{3(2s+1)}{(s^2-2s+2)}$

Solution: (A)

Exercise Problem

1. (A)

Forward path gains, $P_1 = \frac{1}{s}$ and $P_2 = \frac{1}{s^3}$

$\Delta_1 = \Delta_2 = 1$

Loop gains $L_1 = -\frac{1}{s^2}, L_2 = -\frac{1}{s^2}$ and $L_3 = \frac{1}{s^2}$

Transfer function $T(s) = \frac{P_1\Delta_1 + P_1\Delta_2}{1 - L_1 - L_2 - L_3}$

$= \frac{\frac{1}{s} + \frac{1}{s^3}}{1 + \frac{1}{s^2} + \frac{1}{s^2} - \frac{1}{s^2}} = \frac{(s^2+1)}{s(s^2+1)} = \frac{1}{s}$

2. (C)

Between y_1 and y_2

Forward paths,

$P_1 = a$

$P_2 = b$

Loop $\Rightarrow L_1 = c$

$T.F. = \frac{P_1\Delta_1 + P_2\Delta_2}{\Delta} = \frac{a(1-0) + b(1-0)}{1-c}$

$= \frac{a+b}{1-c}$

3. (D)

$\frac{X_2}{X_1} = \frac{5}{\Delta} = \frac{5}{1-(0.5)} = \frac{5}{0.5} = 10$

4. (A)

Using Mason's gain formulae

$\Delta = 1 - \left[-2s^{-1} - 2s^{-2} - 4 - 4s^{-1}\right]$

$\Delta = 1 + \frac{2}{s} + \frac{2}{s^2} + 4 + \frac{4}{s}$

$\Delta = \frac{5s^2 + 6s + 2}{s^2}$

$P_1 = s^{-2}$

$= \frac{1}{s^2}$

$P_2 = s^{-1}$

$= \frac{1}{s}$

$\Delta_1 = 1$

$\Delta_2 = 2$

$\frac{Y(s)}{U(s)} = \frac{\sum P_K\Delta_K}{\Delta}$

$= \frac{\frac{1}{s^2} \times 1 + \frac{1}{s} \times 1}{\frac{5s^2 + 6s + 2}{s^2}}$

$\frac{Y(s)}{u(s)} = \frac{s+1}{5s^2 + 6s + 2}$

5. (C)

Using Mason gain formula,

Transfer function, $G(s) = \frac{C(s)}{U(s)} = \frac{\Sigma P_K\Delta_K}{\Delta}$

Forward paths are:

$P_1 = \frac{h_0}{s^2}, P_2 = \frac{h_1}{s}$

Individual loops are:

$$L_1 = \frac{-a_1}{s}, L_2 = \frac{-a_0}{s^2}$$

Non-touching loops = zero

$\therefore \quad \Delta_1 = 1,$

$$\Delta_1 = 1 - \left(\frac{-a_1}{s}\right)$$

$$= \left(1 + \frac{a_1}{s}\right) = \left(\frac{s + a_1}{s}\right)$$

$$\Delta = 1\left(\frac{-a_1}{s} - \frac{a_0}{s^2}\right) = \frac{s^2 + a_1 s + a_0}{s^2}$$

$$\therefore G(s) = \frac{C(s)}{U(s)} = \frac{P_1\Delta_1 + P_2\Delta_2}{\Delta}$$

$$= \frac{\left(\frac{h_0}{s^2}\right) \times 1 + \left(\frac{h_1}{s}\right) \times \left(\frac{s + a_1}{s}\right)}{\left(\frac{s^2 + a_1 s + a_0}{s^2}\right)}$$

$$= \left[\frac{h_0 + h_1(s + a_1)}{s^2 + a_1 s + a_0}\right]$$

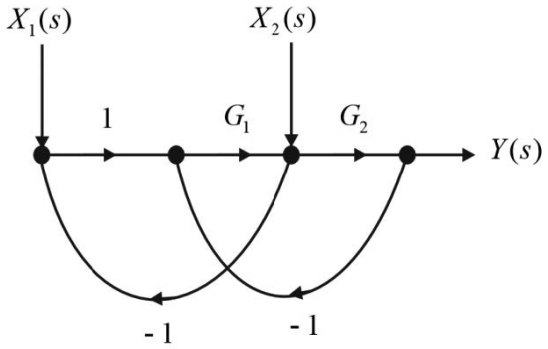
Given, $h_1 = b_1$ and $h_0 = b_0 - b_1 a_1$

Thus, $G(s) = \dfrac{(b_0 - b_1 a_1) + b_1(s + a_1)}{(s^2 + a_1 s + a_0)}$

$$= \left(\frac{b_1 s + b_0}{s^2 + a_1 s + a_0}\right)$$

$$G(s) = \left(\frac{b_1 s + b_0}{s^2 + a_1 s + a_0}\right)$$

6. (B)



$$\left.\frac{Y(s)}{X_2(s)}\right|_{X_1(s)=0} = \frac{G_2}{1 - (-G_1 G_2 - G_1)}$$

$$= \frac{G_2}{1 + G_1(G_2 + 1)}$$

7. (A)

$$\frac{dx_1}{dt} = 2x_1 + x_2 + u \qquad ...(i)$$

$$\frac{dx_2}{dt} = -2x_1 + 4 \qquad ...(ii)$$

From equation (i)

$$sX_1 = 2X_1 + X_2 + u$$

$$sX_2 = -2X_1 + u$$

$$X_2 = \left(-\frac{2X_1 + u}{s}\right)$$

$$sX_1 = 2X_1 + \left(-\frac{2X_1 + u}{s}\right) + 4$$

$$sX_1 - 2X_1 + \frac{2X_1}{s} = \frac{4}{5} + 4$$

$$X_1\left(s - 2 + \frac{2}{s}\right) = u\left(1 + \frac{1}{s}\right)$$

$$\frac{X_1}{4} = \frac{\left(1 + \frac{1}{s}\right)}{\left(s - 2 + \frac{2}{s}\right)}$$

$$\Rightarrow \frac{y}{u} = \frac{3X_1}{u} = \frac{3\left(1 + \frac{1}{s}\right)}{\left(s - 2 + \frac{2}{s}\right)}$$

$$\Rightarrow \frac{y}{u} = \frac{3(s + 1)}{(s^2 - 2s^2 + 2)}$$

4 Mathematical Modelling

Objective

After completing this chapter,the user will be able to understand:

- Mathematical models of various systems
- Force (Torque):Current analogy
- Force (Torque): Voltage analogy

Introduction

- A physical system is a collection of physical objects connected together to serve an objective.
- Examples of a physical system may be cited from the laboratory, industrial plants or utility services - anelectronic amplifier composed of many components, the governing mechanism of a steam turbine, or a communications satellite orbiting the earth are examples of physical systems.
- A more general term system is used to describe a combination of components that may not all be physical, e.g., biological, economic, socio-economic, or management systems.
- The mathematical representation of a physical model through the use of appropriate physical laws is called a mathematical model.

Linear Mode

A mathematical model is linear, if the differential equations describing it have coefficients that are either function only of the independent variable or are constants.

Linear Time-varying Mode

If the coefficients of the describing differential equations are functions of time (the independent variable), then the mathematical model is linear time-varying.

Linear Time-invariant Mode

If the coefficients of the describing differential equations are constants, the model is linear time-invariant. The classical way of modelling a linear time-invariant system is to represent input-output relations between variables.

Mechanical Systems

All mechanical systems are divided into two parts:

1. Translational
2. Rotational

Mechanical Translational System

The translation motion is defined as a motion that takes place along a straight path.

Elements that are used to model mechanical translational systems are mass, spring and damper.

Input = Force (F).

Output = Linear displacement (x) or Linear velocity (v)

1. Mass element:

$$F = M\frac{d^2x}{dt^2}$$

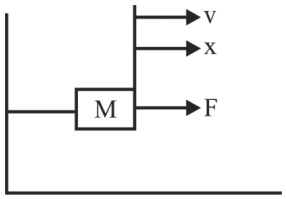


Fig. 4.1

or $F = M\frac{dv}{dt}$

M = Mass (in Kg)

2. Damper or dashpotelement:

$$F = f\frac{d}{dt}(x_1 - x_2) = f\frac{dx}{dt}$$

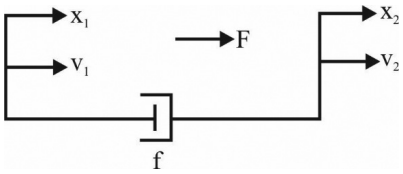


Fig. 4.2

where $x_1 - x_2 = x$

Or $F = f(v_1 - v_2) = fv$

where $v = v_1 - v_2$

f = Viscous frictional coefficient (in N/m/sec)

3. Linear spring element:

$F = K(x_1 - x_2) = Kx$

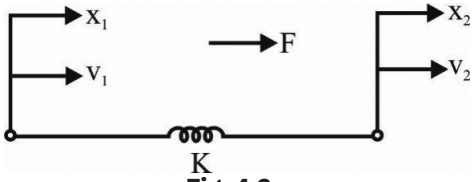


Fig. 4.3

where $x_1 - x_2 = x$

Or $F = K\int(v_1 - v_2) = K\int v$

where $v = v_1 - v_2$

K = Spring constant or stiffness constant (in N/m)

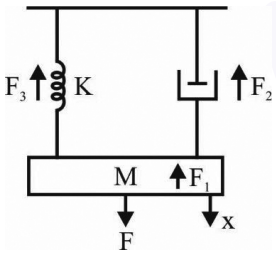


Fig. 4.4

$F = F_1 + F_2 + F_3$

$F = M\frac{d^2x}{dt^2} + f\frac{dx}{dt} + Kx$

Mechanical Rotational System

The motion about a fixed axis is called rotational motion

Input = Torque (τ),

Output = Angular displacement (θ) or Angular velocity (ω)

The three ideal elements used to model rotational systems are inertia, torsional damper and torsional spring.

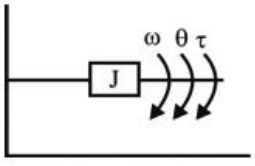


Fig. 4.5

1. Inertial element:

$\tau = J\,d^2\theta / dt^2$

Or $\tau = J\,d\omega / dt$

J = Inertia (Kg – m^2)

2. Torsional damper element:

$\tau = f\frac{d}{dt}(\theta_1 - \theta_2) = f\frac{d\theta}{dt}$,

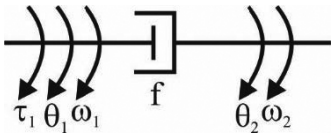


Fig. 4.6

where $\theta = \theta_1 - \theta_2$

Or, $\tau = f(\omega_1 - \omega_2) = f\omega$

where $\omega = \omega_1 - \omega_2$

3. Torsional spring element:

$\tau = K\theta$

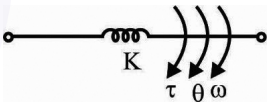


Fig. 4.7

Or $\tau = K\int \omega\,dt$

Consider,

$\tau = \tau_1 + \tau_2 + \tau_3$

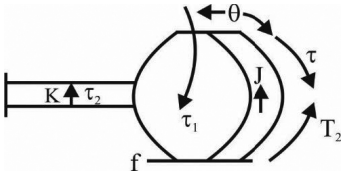


Fig. 4.8

$\tau = J\frac{d^2\theta}{dt^2} + f\frac{d\theta}{dt} + K\theta$ ...(ii)

Analogy

Table 4.1: Analogous Quantities in Force (Torque)-Voltage Analogy (Series Analogy)

If equations of motions of mechanical system are compared with mesh equations, the resulting electric circuit is called as series analogy.

Mechanical translational systems	Mechanical rotational systems	Electrical systems
Force F	Torque T	Voltage e

Mass M	Moment of inertia J	Inductance L
Viscous friction coefficient f	Viscous friction coefficient f	Resistance R
Spring stiffness coefficient K	Tensional or Torsional spring stiffness Coefficient K	Reciprocal of capacitance 1/C
Displacement x	Angular displacement θ	Charge q
Velocity v	Angular velocity ω	Current i

Table 4.2: Analogous Quantities in Force (Torque)-Current Analogy (Parallel Analogy)

If equations of motions of mechanical systems are compared with nodal equations, the resulting electric circuit is called as a parallel analogy.

Mechanical translational systems	Mechanical rotational systems	Electrical systems
Force F	Torque T	Current i
Mass M	Moment of inertia J	Capacitance C
Viscous friction coefficient f	Viscous friction coefficient f	Reciprocal of resistance 1/R
Spring stiffness coefficient K	Tensional or Torsional spring stiffness coefficient K	Reciprocal of inductance 1/L
Displacement x	Angular displacement θ	Magnetic flux linkage λ
Velocity v	Angular velocity ω	Voltage e

Electrical Network

- The resistor, inductor and capacitor are the three basic elements of electrical circuits. These circuits are analysed by the application of Kirchhoff's voltage and current laws.
- We sum voltages around the loop and sum currents at nodes, depending on which technique involves the least effort in algebraic manipulations, and then equate the result to zero.
- From these relationships, we can write the differential equations for the circuit. Then we can take the Laplace transform of the differential equations and finally solve for the transfer function.

Component	Voltage-current	Current-voltage	voltage-charge	Impedance Z(s) = V(s)/I(s)	Admittance Y(s) = I(s)/V(s)
Capacitor	$v(t)=\frac{1}{C}\int_0^t i(\tau)d\tau$	$i(t)=C\frac{dv(t)}{dt}$	$v(t)=\frac{1}{C}q(t)$	$\frac{1}{Cs}$	Cs
Resistor	$v(t)=Ri(t)$	$i(t)=\frac{1}{R}v(t)$	$v(t)=R\frac{dq(t)}{dt}$	R	$\frac{1}{R}=G$
Inductor	$v(t)=L\frac{di(t)}{dt}$	$i(t)=\frac{1}{L}\int_0^t v(\tau)d\tau$	$v(t)=L\frac{d^2q(t)}{dt^2}$	Ls	$\frac{1}{Ls}$

Table 4.3: Voltage-Current, Voltage-Charge, and Impedance Relationship for Capacitors, Resistors and Inductors

Note: The following set of symbols and units are used throughout this book: v(t) – V (volts), i(t) = A (amps), q(t) – Q (coulombs), C – F (farads), $R-\Omega$ (ohms), L – H (henries).

Example: Find the transfer function relating the capacitor voltage, $V_c(s)$, to the input voltage, V(s) in figure.

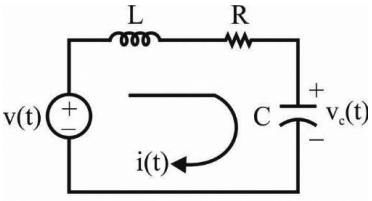


Fig. 4.9 RLC Network

Solution: In any problem, the designer must first decide what input and output should be. In this network, several variables could have been chosen to be as voltage or current. The problem statement, however, is clear in this case: We have to treat the capacitor voltage as the output and the applied voltage as the input.

Summing the voltages around the loop, assuming zero initial conditions, yields the integro-differential equation for this network as

$$L\frac{di(t)}{dt}+Ri(t)+\frac{1}{C}\int_0^t i(\tau)d\tau=v(t) \qquad (1)$$

Changing variables from current to charge using i(t) = dq(t)/dt yields

$$L\frac{d^2q(t)}{dt^2}+R\frac{dq(t)}{dt}+\frac{1}{C}q(t)=v(t) \qquad (2)$$

From the voltage-charge relationship for a capacitor in Table

$$q(t)=Cv_c(t) \qquad (3)$$

Substituting equation (3) into equation (2) yields

$$LC\frac{d^2v_c(t)}{dt^2}+RC\frac{dv_c(t)}{dt}+v_c(t)=v(t) \qquad (4)$$

Taking the Laplace transform assuming zero initial conditions, rearranging terms, and simplifying yields.

$$\left(LCs^2+RCs+1\right)V_c(s)=V(s) \qquad (5)$$

Solving for the transfer function, $V_c(s)/V(s)$, we obtain

$$\frac{V_c(s)}{V(s)}=\frac{1/LC}{s^2+\frac{R}{L}s+\frac{1}{LC}} \qquad ...(6)$$

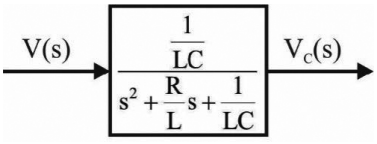


Fig. 4.10 Block Diagram of series RLC Electrical Network

Series RLC Circuit

Applying KVL,

$$iR+L\frac{di}{dt}+\frac{1}{C}\int i\,dt=v_{in}$$

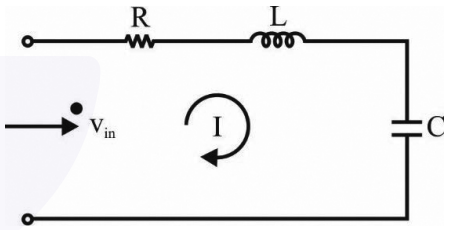


Fig. 4.11

But, $i=\frac{dq}{dt}$

Or, $q=\int i\,dt$

$$L\frac{d^2q}{dt^2}+R\frac{dq}{dt}+\frac{q}{C}=v_{in}$$

Parallel RLC Circuit

Applying KVL,

$$\frac{V}{R}+\frac{1}{L}\int v\,dt+C\frac{dv}{dt}=i$$

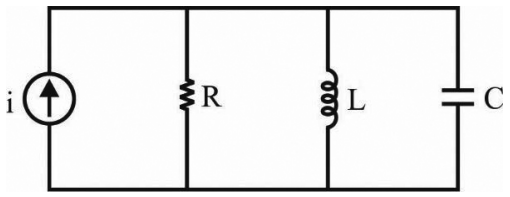


Fig. 4.12

Where v = node voltage

$$v=\frac{d\phi}{dt}$$

Where ϕ = *Magnetic flux*

$$C\frac{d^2\phi}{dt^2}+\frac{1}{R}\frac{d\phi}{dt}+\frac{\phi}{L}=i$$

1. **Duality:**

   $C \rightarrow L$,

   $R \rightarrow \frac{1}{R}$,

   $\phi \rightarrow q$

2. **Series** analogue**:**

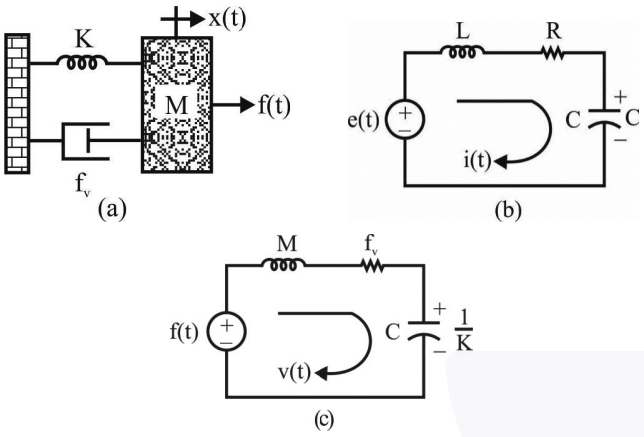


Fig. 4.13

   Mass = M → Inductor = M henries

   Viscous damper = f_v → Resistor = f_v ohms

   Spring = K → Capacitor = $\frac{1}{K}$ *farads*

   Applied force = f(t) → applied voltage = f(t)

   Velocity = v(t) → mesh current = v(t)

3. **Parallel analogy:**

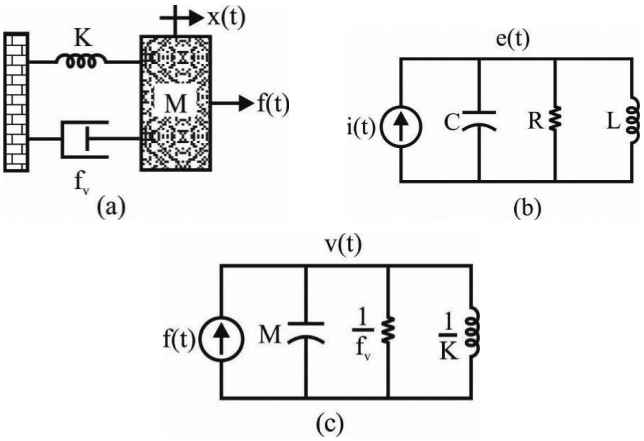


Fig. 4.14

Mass = M → capacitor = M farads

Viscous damper = f_v → resistor = $\frac{1}{f_v}$ ohms

Spring = K → inductor = $\frac{1}{K}$ *henries*

Applied force = f(t) → current source = f(t)

Velocity = v(t) → node voltage = v(t)

Nodal Method for Writing Differential Equations of Complex Mechanical System

1. Number of nodes = Number of displacements.
2. Take an additional node which is the reference node.
3. Connect the mass and inertial mass elements always between the principle node and reference node.
4. Connect the spring and damping elements either between the principle nodes or between the principle node and reference node, depending on their position.
5. Obtain the nodal diagram and write the describing differential equations at each node.

Representation of "Mass-Damper" System into Series-parallel Network

1. Ground is taken as reference
2. Every mass has its displacement
3. Every displacement is represented by a "Node"
4. Displacement of every mass, displacement of every friction (damping) and displacement of spring is taken as independent displacement.
5. Every mass is connected between the reference node and the node corresponding to its displacement.
6. The elements connected between ground and mass in the system are always connected between the reference node and displacement corresponding to mass.
7. "The elements connected between two masses are always connected between the nodes corresponding to the displacement of those masses".
8. "A force acting on a mass is always connected between the reference node and the node corresponding to the displacement of mass".

Example:

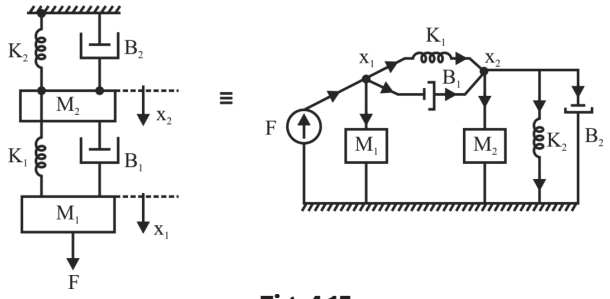


Fig. 4.15

Node x_1:

$$F = M_1\frac{d^2x_1}{dt^2} + k_1\left(x_1 - x_2\right) + B_1\frac{d}{dt}\left(x_1 - x_2\right) \quad (1)$$

$$K_1\left(x_1 - x_2\right) + B_1\frac{d}{dt}\left(x_1 - x_2\right) = M_2\frac{d^2x_2}{dt^2} + k_2x_2 + B_2\frac{dx_2}{dt} \quad (2)$$

Points To Remember

- One mass element ⇒ Order of differential equation = 2
  Two mass elements ⇒ Order of differential equation = 4
  "n" mass elements ⇒ Order of differential equation = 2n
- A point or node between damper and spring must be taken as a displacement

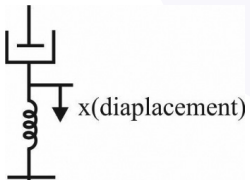


Fig. 4.16

- Each mass must be taken as a displacement.

Gear Train

A gear train is a mechanical device which transfers energy from one part of the system to other parts without any loss.

N = number of teeth on the circumference of the gear wheel

r = radius of the gear wheel (m)

τ = torque (N-m)

θ = angular displacement (radians)

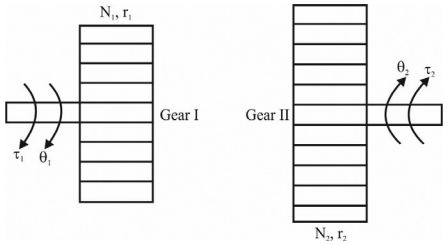


Fig. 4.17

For two wheels $\frac{N_1}{N_2} = \frac{\tau_1}{\tau_2} = \frac{r_1}{r_2} = \frac{\theta_2}{\theta_1} = \frac{\omega_2}{\omega_1} = \frac{\alpha_2}{\alpha_1}$

For N wheel connected in series

$$\frac{N_x}{N_y} = \frac{\tau_x}{\tau_y} = \frac{r_x}{r_y} = \frac{\theta_y}{\theta_x} = \frac{\omega_y}{\omega_x} = \frac{\alpha_y}{\alpha_x}$$

Examples:

1. Consider a mechanical system, as shown in the figure. Write the force equations and draw a fully labelled state diagram.

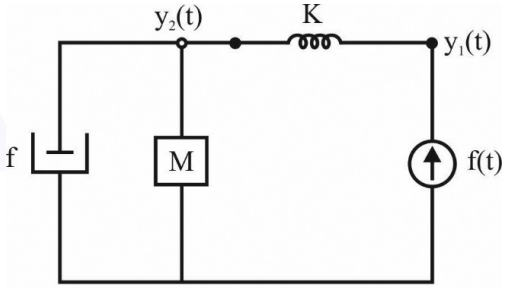


Fig. 4.18 Reference Node

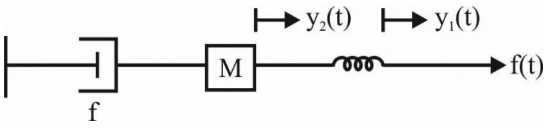


Fig. 4.19

Solution: Equivalent diagram will be

At node $y_2(t)$

$$M\frac{d^2y}{dt^2} + f\frac{dy_2}{dt} + K\left(y_2 - y_1\right) = 0$$

And $f\left(t\right) = K\left(y_1 - y_2\right)$

So, $M\frac{d^2y_2}{dt^2} + f\frac{dy_2}{dt} - f\left(t\right) = 0$

State diagram using integrator is

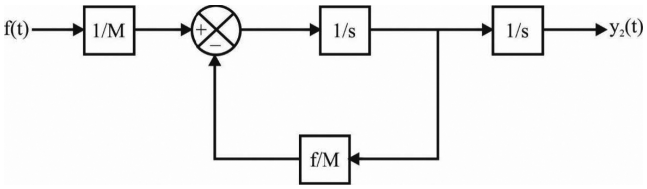


Fig. 4.20

2. Obtain the mathematical equations and its corresponding mechanical circuit diagram of the following mechanical system.

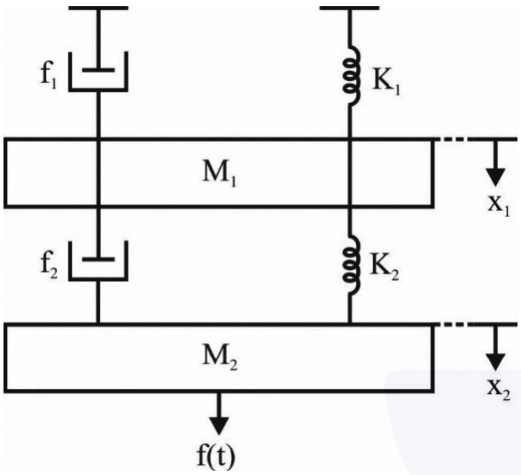


Fig. 4.21 Mechanical System

Solution: At node x_1:

$$f_1\frac{dx_1}{dt}+K_1x_1+M_1\frac{d^2x_1}{dt^2}+K_2\left(x_1-x_2\right)$$
$$+f_2\frac{d\left(x_1-x_2\right)}{dt}=0$$

At node x_2:

$$M_2\frac{d^2x_2}{dt^2}+K_2\left(x_2-x_1\right)+f_2\frac{d\left(x_2-x_1\right)}{dt}=f\left(t\right)$$

Now, mechanical circuit diagrams can be drawn either from a given system diagram or from these system equations as well.

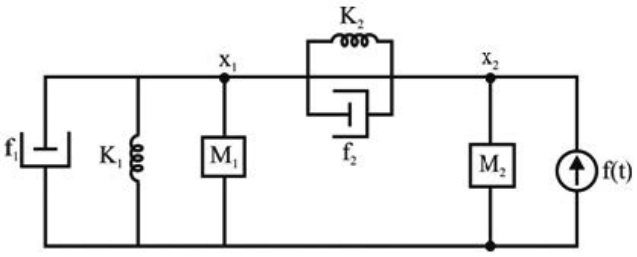


Fig. 4.22 Mechanical Circuit Diagram

3. Find the transfer function relating displacements y and x for the mechanical system of the given figure.

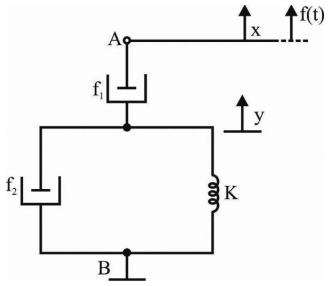


Fig. 4.23

Solution: From the mechanical circuit diagram, the equations for the system are

$$f_2\frac{dy}{dt}+Ky+f_1\frac{d}{dt}\left(y-x\right)=0 \quad (1)$$

$$f_1\frac{d}{dt}\left(x-y\right)=f\left(t\right) \quad (2)$$

Assuming initial conditions as zero and taking Laplace transform on both sides of equation (1),

$$f_2sY\left(s\right)+KY\left(s\right)+f_1s\left[Y\left(s\right)-X\left(s\right)\right]=0 \quad (3)$$

From equation (3), the required transfer function is

$$\frac{Y\left(s\right)}{X\left(s\right)}=\frac{f_1s}{\left[\left(f_1+f_2\right)s+K\right]}=\frac{T_1s}{\left(T_2s+1\right)}$$

where $T_1=\frac{f_1}{K}$ and $T_2=\frac{\left(f_1+f_2\right)}{K}$

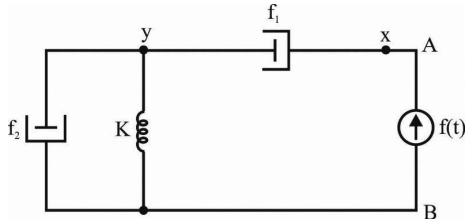


Fig. 4.24

4. Convert the given mechanical circuit diagram into a corresponding electrical analogous system, driven by the voltage source and current source.

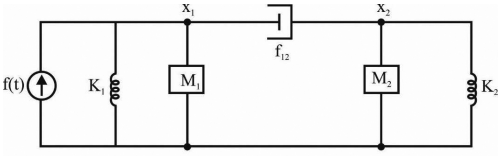


Fig. 4.25 Mechanical Circuit Diagram

Solution:

At node x_1: $f\left(t\right)=M_1\frac{d^2x_1}{dt^2}+f_{12}\frac{d}{dt}\left(x_1-x_2\right)+K_1x_1$

At node x_2: $0 = M_2 \frac{d^2x_2}{dt^2} + f_{12}\frac{d}{dt}(x_2 - x_1) + K_2 x_2$

Now, using F-V analogy:

$F \rightarrow V$

$M \rightarrow L$

$f \rightarrow R$

$K \rightarrow 1/C$

$v \rightarrow i$

$x \rightarrow q$

Hence, the corresponding electrical system (driven by voltage source) will have the following system equations

$V = L_1 \frac{di_1}{dt} + R_{12}(i_1 - i_2) + \frac{1}{C_1}\int i_1 dt$ (First loop equation)

And $0 = L_2 \frac{di_2}{dt} + R_{12}(i_2 - i_1) + \frac{1}{C_2}\int i_2 dt$

(Second loop equation)

Hence, the (analogous) electrical circuit diagram will be

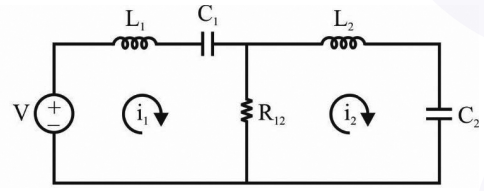


Fig. 4.26 Electrical Circuit Diagram, Driven by Voltage Source

Now, using F-I analogy:

$F \rightarrow I$

$M \rightarrow C$

$f \rightarrow 1/R$

$K \rightarrow 1/L$

$v \rightarrow V$

$x \rightarrow \phi$

Hence, the corresponding electrical system (driven by the current source) will have the following system equations:

$I = C_1 \frac{dv_1}{dt} + \frac{V_1 - V_2}{R_{12}} + \frac{1}{L_1}\int V_1 dt$ (First Nodal equation)

And $0 = C_2 \frac{dv_2}{dt} + \frac{V_2 - V_1}{R_{12}} + \frac{1}{L_2}\int V_2 dt$ (Second Nodal equation)

Hence, the (analogous) electrical circuit diagram will be

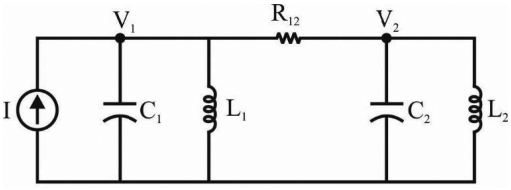


Fig. 4.27 Electrical Circuit Diagram, Driven by current Source

5. Derive an expression for the transfer function $\frac{X_0(s)}{X_i(s)}$ for the mechanical system given below:

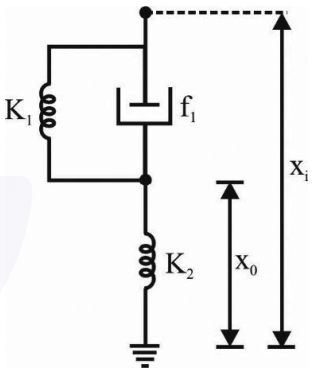


Fig. 4.28

Solution:

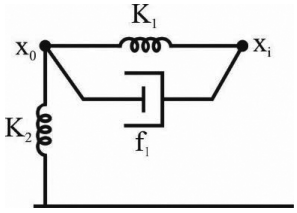


Fig. 4.29

t node x_0

$K_2 x_0 + K_1(x_0 - x_i) + f_1\frac{d}{dt}(x_0 - x_i) = 0$

Using Laplace transform,

$K_2 X_0(s) + K_1(X_0(s) - X_i(s))$

$+ f_1 s(X_0(s) - X_i(s)) = 0$

On solving, $\frac{X_0(s)}{X_i(s)} = \frac{f_1 s + K_1}{f_1 s + K_1 + K_2}$

6. A mechanical system is shown in the figure below. The system is set into motion by applying a unit impulse force $\delta(t)$. Assuming that the system is initially

at rest and ignoring friction, what is the displacement x(t) of mass?

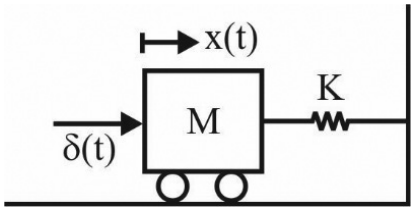


Fig. 4.30

(A) $\frac{1}{\sqrt{K}}\exp\left(-M.t\right)$

(B) $\frac{1}{\sqrt{MK}}\sin\left(t\right)$

(C) $\frac{1}{\sqrt{MK}}\sin\left(\sqrt{\frac{K}{M}}.t\right)$

(D) $\frac{1}{\sqrt{MK}}\left(\sqrt{\frac{K}{M}}.t\right)$

Solution: (C)

$$\delta\left(t\right)=\frac{Md^2x\left(t\right)}{dt^2}+Kx\left(t\right)$$

Taking Laplace transform

$$1=Ms^2X\left(s\right)+K\left[X\left(s\right)\right]$$

$$X\left(s\right)=\frac{1}{Ms^2+K}=\frac{1}{M\left[s^2+\frac{K}{M}\right]}$$

$$x\left(t\right)=\frac{1}{\sqrt{MK}}\sin\left(\sqrt{\frac{K}{M}}.t\right)$$

Exercise Problems

1. A torque of T Nm is applied to a shaft having a moment of inertia J, and the coefficient of viscous friction of f produces an angular twist of θ radians. Obtain the transfer function relating θ and T.

Solution:

$$\frac{\theta\left(s\right)}{T\left(s\right)}=\frac{1}{s\left(Js+f\right)}$$

2. In the figure shown stiffness constant is K, the viscous friction coefficient is B, the mass is M, and the system output motion is x(t), corresponding to the input force f(t). Find the following parameters related to the given system.

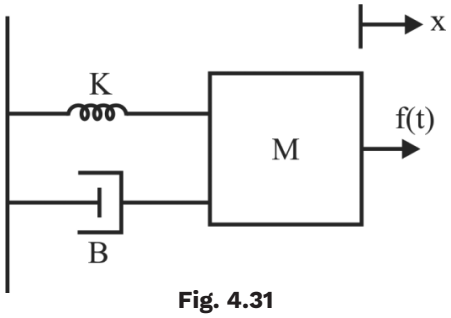


Fig. 4.31

a) Time constant

b) Damping coefficient

c) Natural frequency of oscillations

Solution:

a) $T=\frac{2M}{B}$

b) $\xi=\frac{B}{2\sqrt{MK}}$

c) $\omega_n=\sqrt{\frac{K}{M}}$

3. Consider the following mechanical system shown in the diagram:

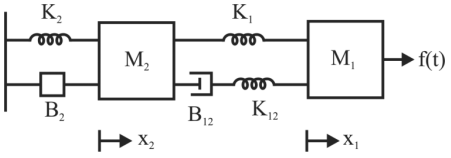


Fig. 4.32

Draw the circuit which shows the correct force-current analogous electrical circuit for the mechanical diagram shown above?

Solution:

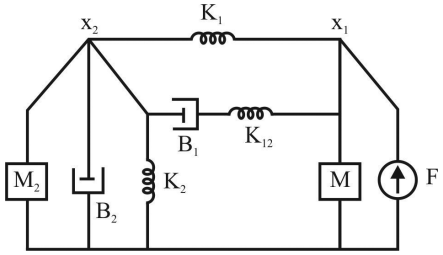
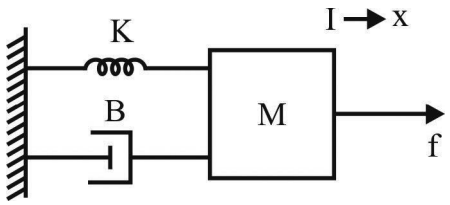


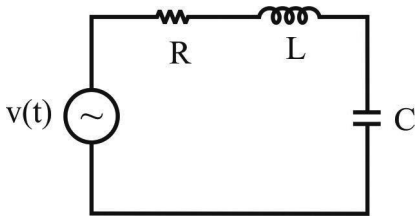
Fig. 4.33

4. Consider the following systems:

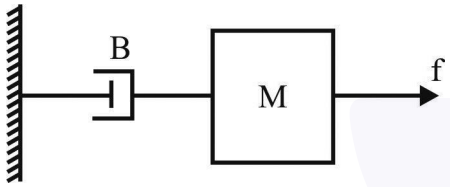
I.



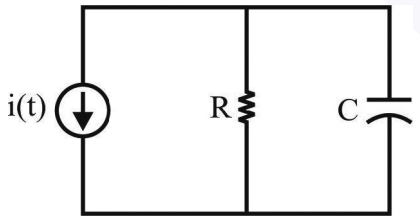
II.



III.



IV.



Which of these systems can be modelled by the differential equation, **[ESE-1998]**

$$a_2\frac{d^2y(t)}{dt^2}+a_1\frac{dy(t)}{dt}+a_0y(t)=x(t)\,?$$

(A) a and b (B) a and c

(C) b and d (D) a, b and d

Solution: (A)

5. Match List-I (Functional components) with List-II (Devices) and select the correct answer: **[ESE-2000]**

List-I

A. Error detector

B. Servomotor

C. Amplifier

D. Feedback

List-II

I. Three-phase FHP induction motor

II. A pair of synchronous transmitters and control transformer

III. Tachogenerator

IV. Amplidyne

V. Missing

Codes:

	A	B	C	D
(A)	II.	IV.	I.	v.
(B)	IV.	II.	V.	III.
(C)	II.	IV.	V.	III.
(D)	I.	II.	III.	V.

Solution: (C)

Exercise Problems

1.

Solution: $\left(\frac{\theta(s)}{T(s)}=\frac{1}{s(Js+f)}\right)$

The equation for the system is given by

$$T=J\frac{d^2\theta}{dt^2}+f\frac{d\theta}{dt} \qquad \ldots(i)$$

Assuming the initial condition as zero and taking the Laplace transformation on both sides of equation (i), the following equation is obtained,

$$T(s)=Js^2\theta(s)+fs\theta(s)$$

Or $T(s)=s(Js+f)+\theta(s)$ $\ldots(ii)$

From equation (ii), the required transfer function is obtained below.

$$\frac{\theta(s)}{T(s)}=\frac{1}{S(Js+f)}$$

2.

Solution: $f(t)=\frac{Md^2x}{dt^2}+B\frac{dx}{dt}+Kx(t)$

Applying Laplace transform on both sides, with zero initial condition

$$F(s)=Ms^2X(s)+BsX(s)+KX(s)$$

$$\frac{X(s)}{F(s)} = \frac{1}{Ms^2 + Bs + K}$$

Characteristic equation is

$$Ms^2 + Bs + K = 0$$

$$s^2 + \frac{B}{M} + \frac{K}{M} = 0$$

Comparing with $s^2 + 2\xi\omega_n s + \omega_n^2 = 0$

$$2\xi\omega_n = \frac{B}{M}$$

$$\xi = \frac{B}{2\sqrt{MK}} \qquad \omega_n = \sqrt{\frac{K}{M}}$$

Time constant $T = \frac{1}{\xi\omega_n} = \frac{1}{B} \times 2M$

$$T = \frac{2M}{B}$$

3.

Solution:

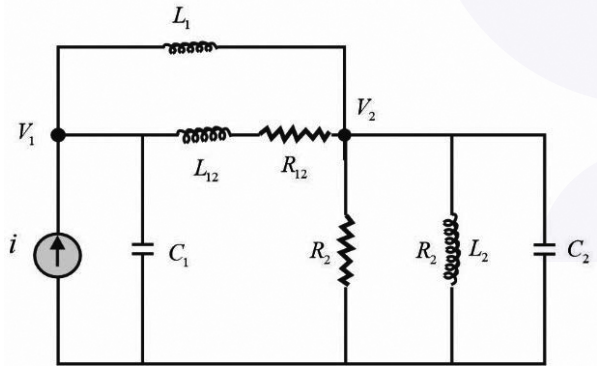


Fig. 4.34

Nodal diagram

Mechanical system	Force current analog
Mass(M)	Capacitance (C)
Damper(B)	Conductance(1/R)
Spring coefficient	Inductance current(1/L)
Force(K)	

Mechanical network is shown below:

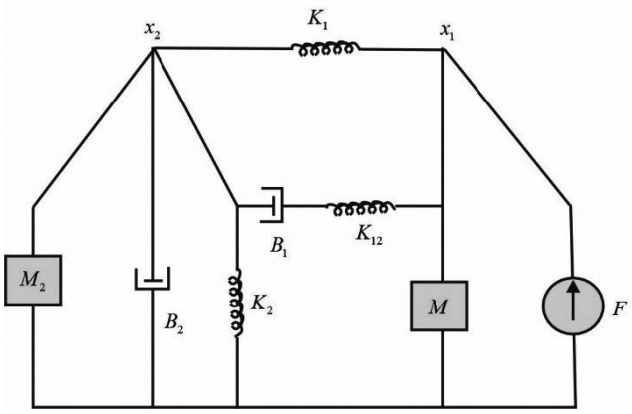


Fig. 4.35

4. (A)

This is a second order differential equation which means that there must be present all the three components in the system, i.e., either R, L and C or K,B and M.

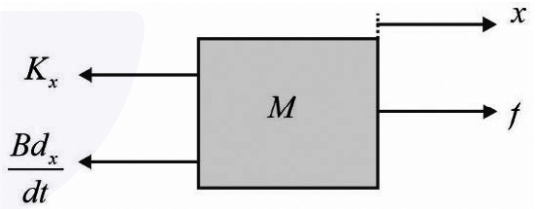


Fig. 4.36 Free Body of M in Diagram

In Figure 1,

$$f - Kx - B\frac{dx}{dt} = M\frac{d^2x}{dt^2}$$

Or $M\frac{d^2x}{dt^2} + B\frac{dx}{dt} + Kx = f$ ...(i)

In figure 2,

$$v(t) = Ri + \frac{Ldi}{dt} + \frac{q}{c}$$

Or $v(t) = \frac{Ld^2q}{dt^2} + R\frac{dq}{dt} + \frac{q}{c}$ ...(ii)

Both the equation are symmetric to the given equation.

5. (C)